UP
Universidad de Palermo

Diplomatura en Inteligencia Artificial 2025

Prof. Carlos A. Giménez

Módulo 2 – Machine Learning

Bloque Temático 2

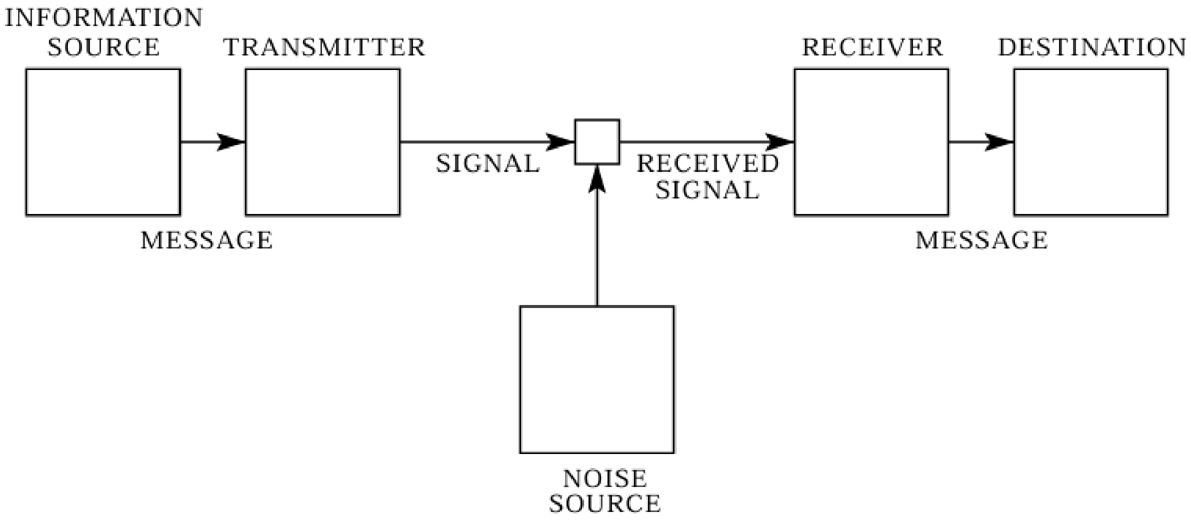
- Intro a Redes Neuronales.
- Feedfoward.

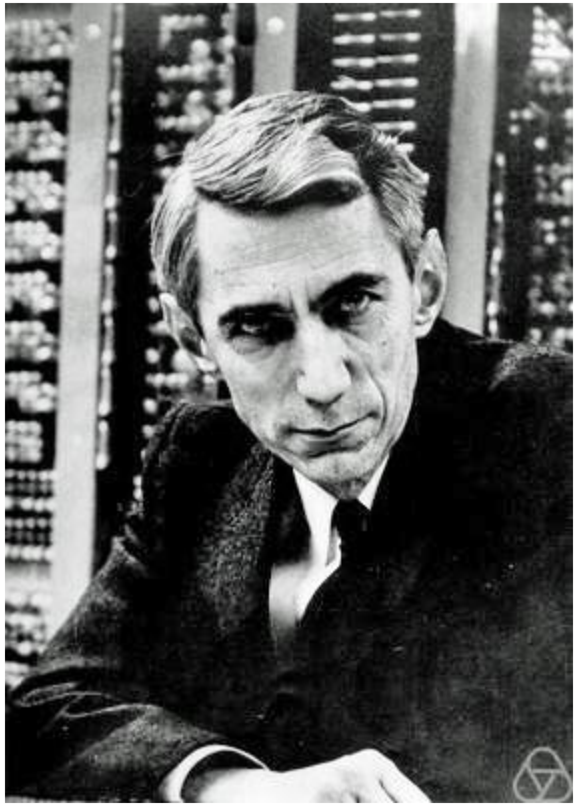
A visual introduction to machine learning

Entropía

Concepto

Claude Elwood Shannon (30 de abril de 1916 - 24 de febrero de 2001) fue un matemático, ingeniero eléctrico y criptógrafo estadounidense recordado como «el padre de la teoría de la información».[1][2]

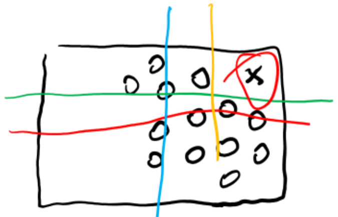




wiki

Entropía

Clasificación



32

$$\frac{32}{2} = 16 \rightarrow \frac{16}{2} \rightarrow 8 \rightarrow \frac{8}{2} = 4 \rightarrow \frac{4}{2} \rightarrow 2 \rightarrow \frac{2}{2} = 1$$

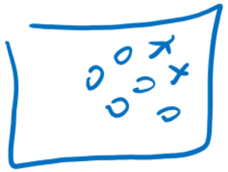
1 2 3 4 5

$$\boxed{\log_2 32 = 5}$$

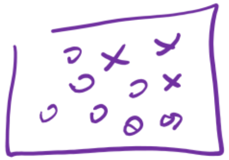
bit

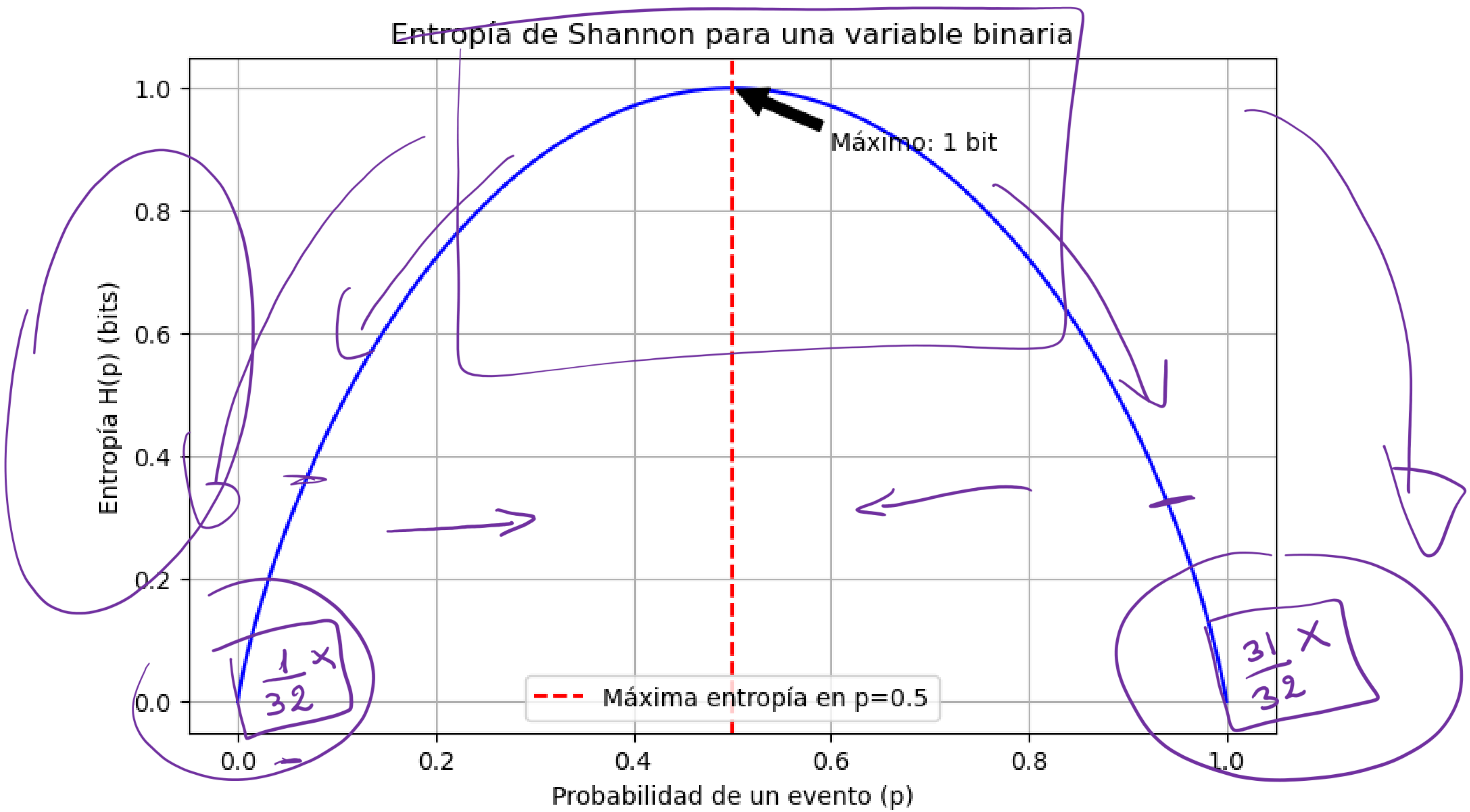
$$\boxed{-p_i \log_2 p_i} \rightarrow p = \frac{1}{32}$$

$\frac{2}{32}$



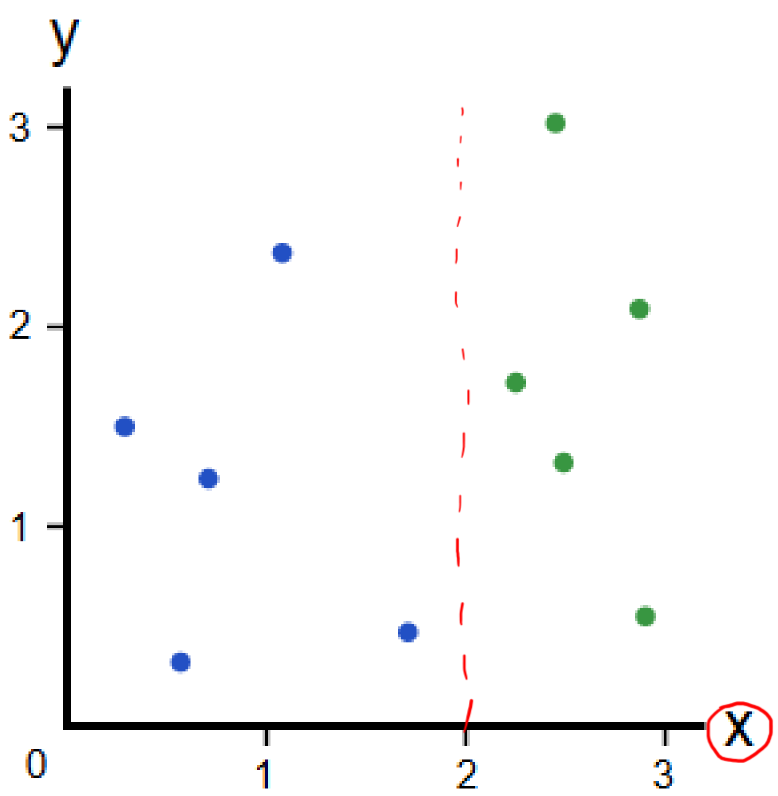
$\frac{3}{32}$




Entropía de Shannon para una variable binaria
Máximo: 1 bit
Máxima entropía en p=0.5
Entropía H(p) (bits)
Probabilidad de un evento (p)
1/32 x
31/32 x

Impureza de GINI
Clasificación

$$G = \sum_{i=1}^{k} p_i * (1 - p_i)$$



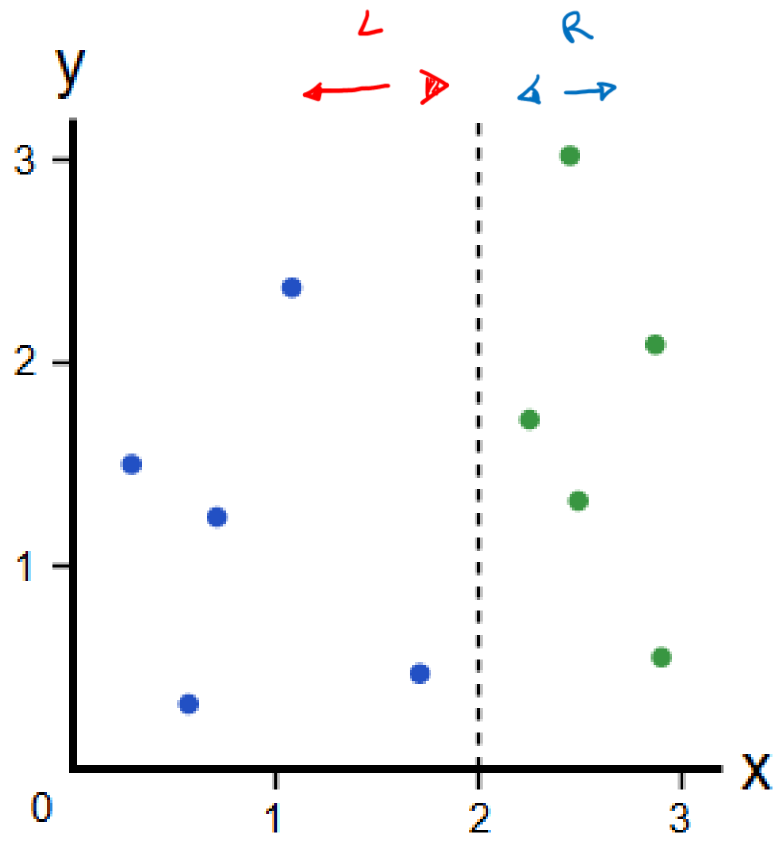
P_A $\frac{5}{10} = 0,5$

P_V $\frac{5}{10} = 0,5$

Si $X = 2$

Impureza de GINI

Clasificación



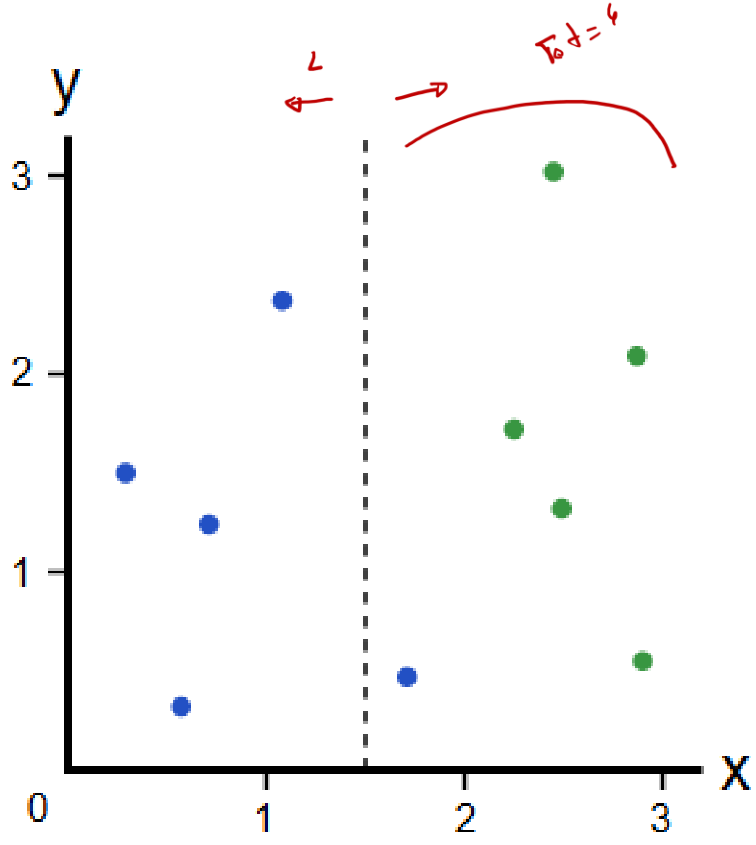
$$G = \sum_{i}^{K} P_i\,(1-P_i) = \underbrace{P_A\cdot(1-P_A)}_{\boxed{\text{AZUL}}} + \underbrace{P_V\cdot(1-P_V)}_{\boxed{\text{VERDE}}}$$

$$G_L = \frac{5}{5}\left(\underbrace{1-\frac{5}{5}}_{=0}\right) + \frac{0}{5}\cdot\cancel{\left(1-\frac{0}{5}\right)} = 0$$

$$G_R = \underbrace{\frac{0}{5}\left(1-\frac{0}{5}\right)}_{=0} + \frac{5}{5}\underbrace{\left(1-\frac{5}{5}\right)}_{=0} = 0$$

Impureza de GINI

Clasificación



Az Vcrd

$$G_L = \frac{4}{4}\left(1-\frac{4}{4}\right) + \frac{0}{4}\left(1-\frac{0}{4}\right) = 0 \quad \left.\right] 4$$

$$G_R = \frac{1}{6}\left(1-\frac{1}{6}\right) + \frac{5}{6}\left(1-\frac{5}{6}\right) = \quad \left.\right] 6$$

$$0{,}138 + 0{,}138 = 0{,}276$$

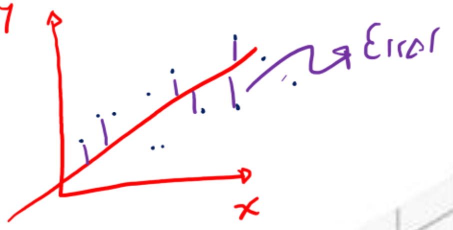
$$G_G = 1 - \left(\frac{4}{10}\cdot G_L + \frac{6}{10}\cdot G_R\right) = 1 - \underbrace{\left[\frac{4}{10}(0) + \frac{6}{10}\cdot(0{,}276)\right]}_{0{,}1656} =$$

$$G_G = 1 - 0{,}1656 = \boxed{0{,}8344}$$

$$G_G \to 1$$

Log Loss

Clasificación



$y = \overset{?}{a}x + \overset{?}{b}$

OLS

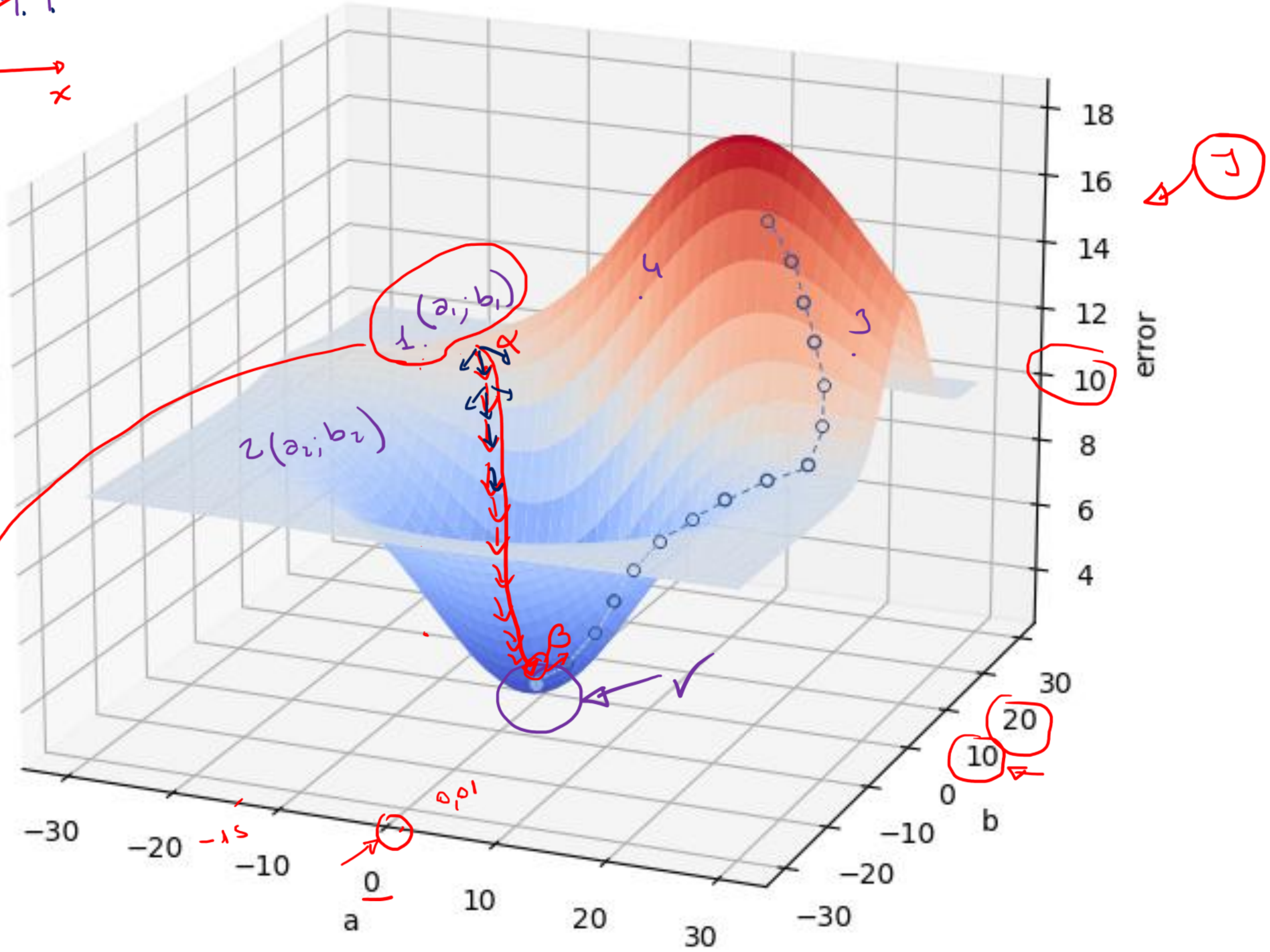
$y' - y = \text{Error}$

$$g(z) = \frac{1}{1 + e^{-\overbrace{(ax+b)}^{z}}}$$

Logit

$b = ?$

$a = ?$



$$g(x) = \frac{1}{1 + e^{(-15x+20)}}$$

$g'(x) - g(x) = 10$

Log Loss
Clasificación

$$f_{w,b}(x) = \frac{1}{1 + e^{-(w+bx)}}$$

logit

Función de pérdida:

$$\underline{L(w,b)} = -\underset{1,\ 0}{\overset{\text{Real}}{y}} \log\left(\underbrace{\overbrace{f_{w,b}(x)}^{\text{Calc.}}}_{0,\ldots,1}\right) - (1 - \overbrace{y}^{\text{Real}})\log(1 - \overbrace{f_{w,b}}^{\text{Calc.}}(x))$$

Función de costo:

$$\underline{J(w,b)} = \frac{1}{m}\sum_{i=1}^{m} L(w,b)$$

Log Loss

Clasificación

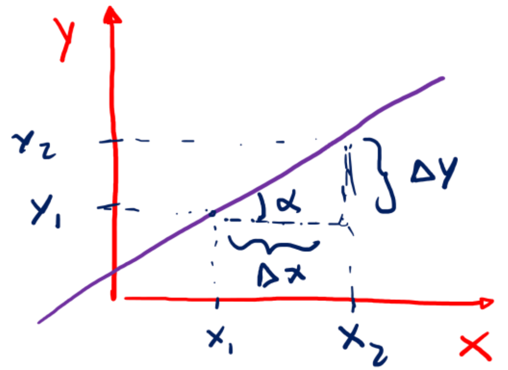
Función de pérdida:

$$f_{w,b}(x) = \frac{1}{1 + e^{-(w+bx)}}$$

$$L(w,b) = -y \log\left(f_{w,b}(x)\right) - (1-y)\log(1 - f_{\mathrm{w,b}}(\mathrm{x}))$$

Función de costo:

$$J(w,b) = \frac{1}{m}\sum_{i=1}^{m} L(w,b)$$



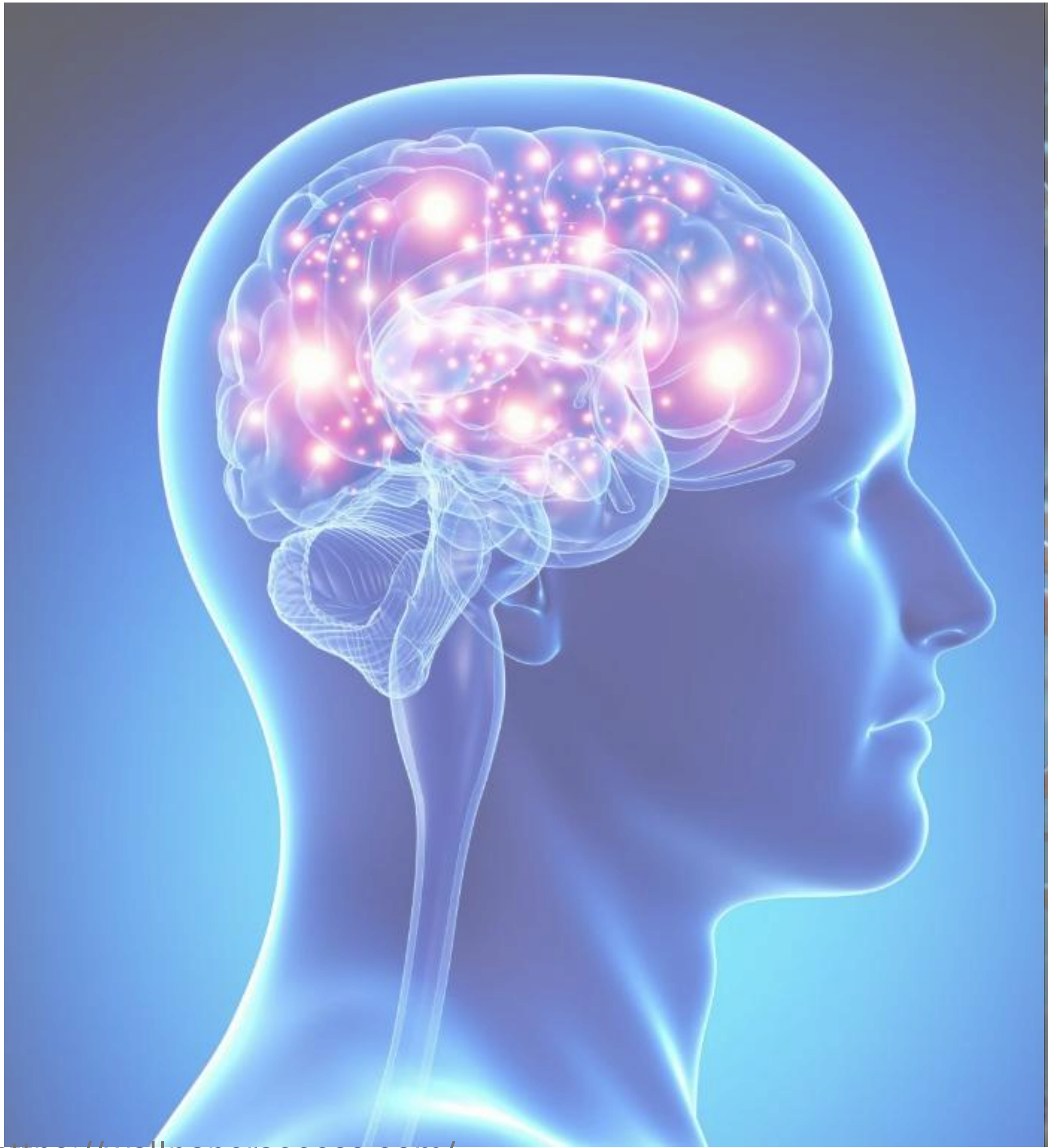
$$\overset{+}{\downarrow} \quad \frac{\Delta y}{\Delta x} \sim \frac{\partial y}{\partial x}$$

Debemos encontrar w, b que minimice J(w,b):

Aplicamos gradiente:

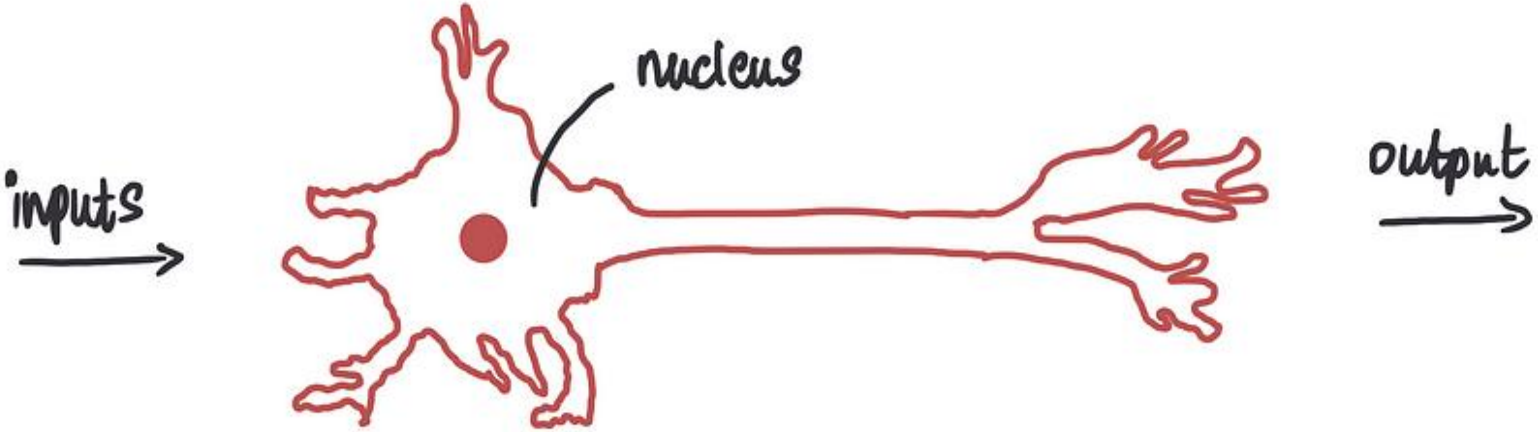
$$\frac{\partial}{\partial b} J(w,b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x_i) - y_i\right) x_i \quad 1$$

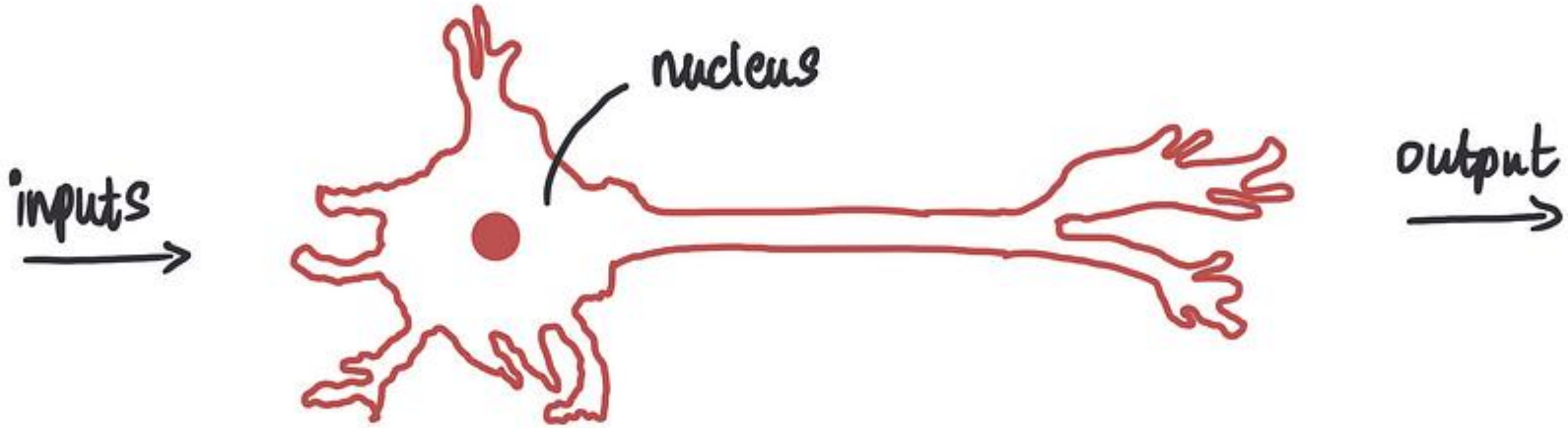
$$\frac{\partial}{\partial w} J(w,b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x_i) - y_i\right) \quad 2$$

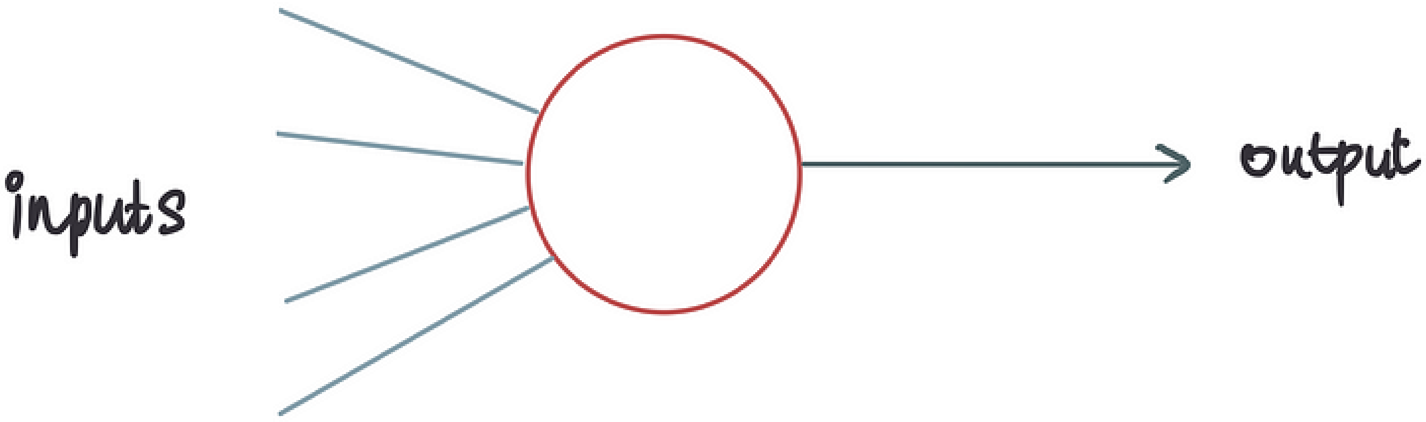



https://wallpaperaccess.com/

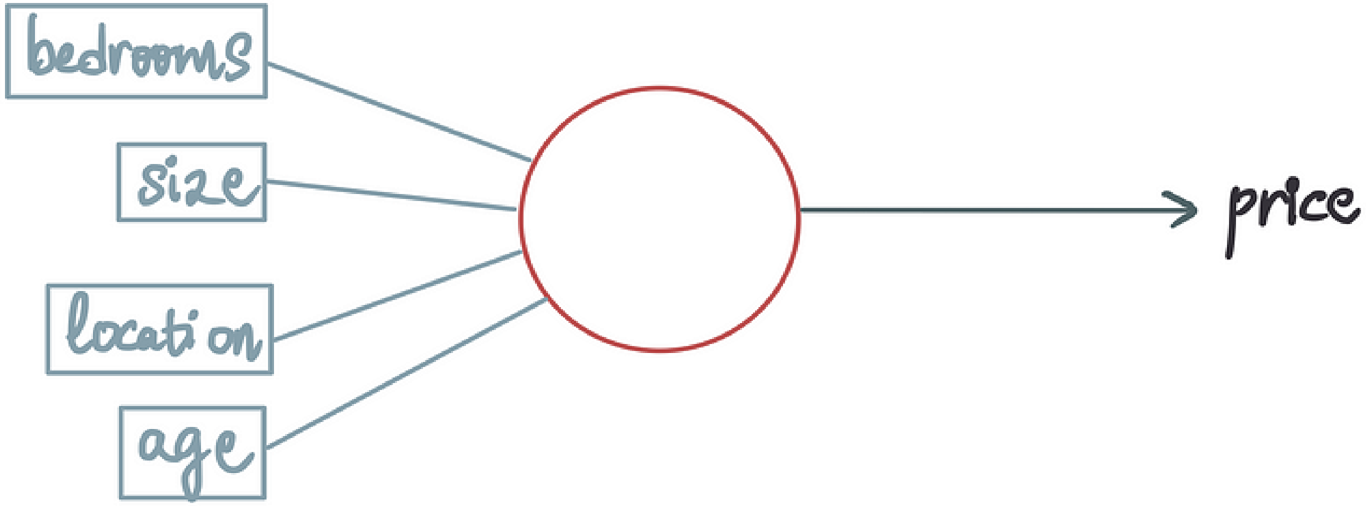

Adobe Stock | #745295183

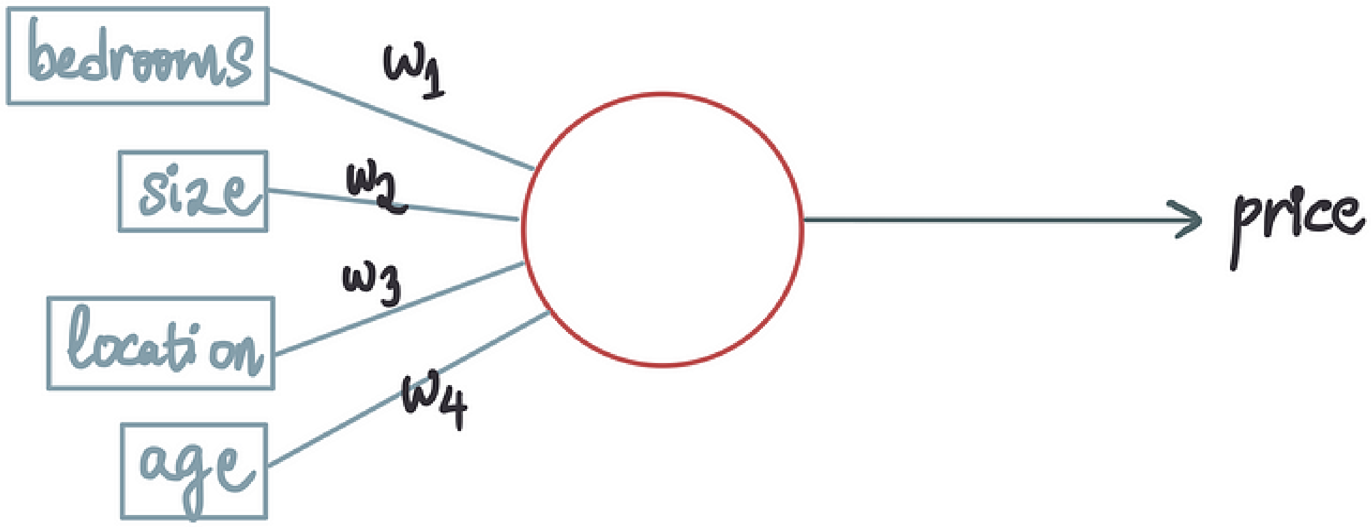

nucleus
inputs
output


nucleus
inputs
output

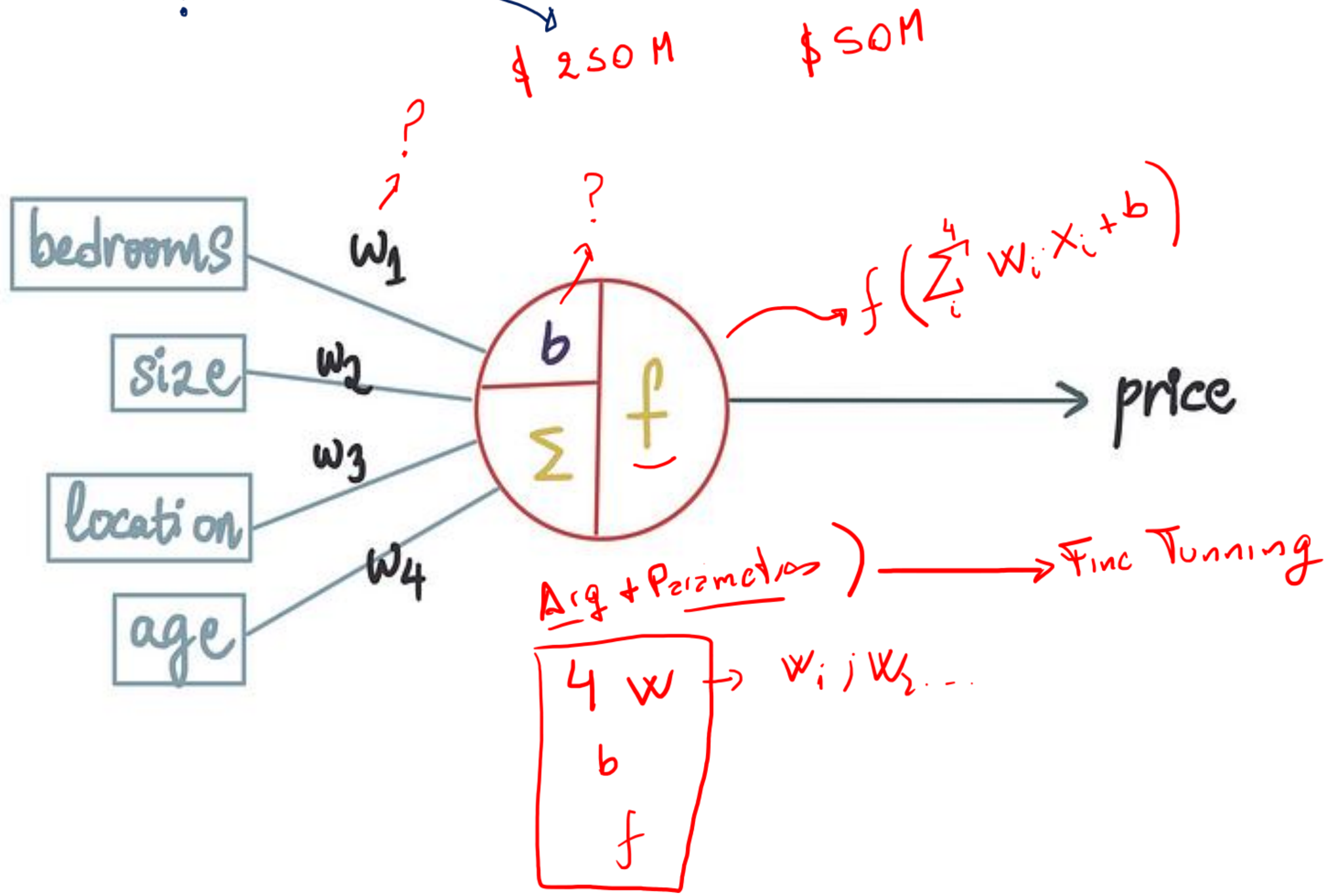

inputs
output

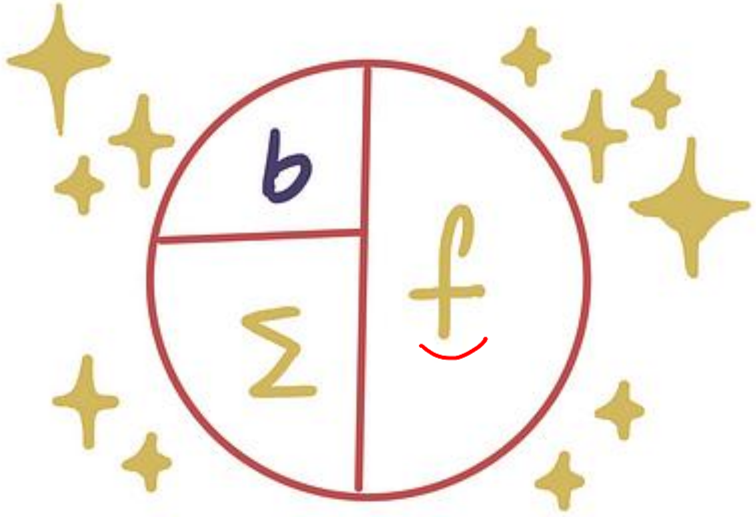
bedrooms	size	location	age	price
4	2680	2	8	1.75M
5	700	1	27	750K
3	2360	1	28	1.2M
5	1280	2	98	1.05M
5	3480	3	21	2.3M

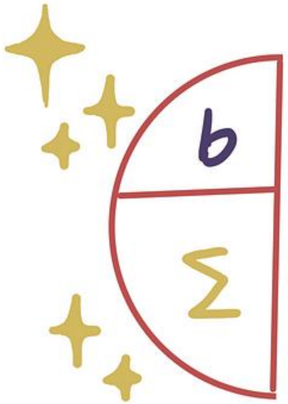

bedrooms
size
location
age
price

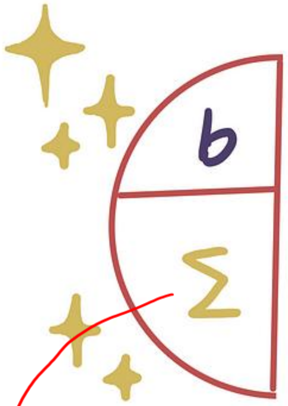

bedrooms
size
location
age
w_1
w_2
w_3
w_4
price

Dimensions → $\underbrace{w_1; w_2; w_3; w_4; b;}_{?}$ Error



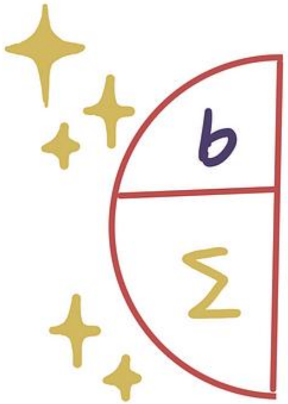

b
Σ
f


b
Σ



Suma

$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age} + b$$



$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age}$$

$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age} + \underbrace{b}_{\text{bias}}$$

magic, part 1

$$w_1 \cdot \text{bedrooms} + w_2 \cdot \text{size} + w_3 \cdot \text{location} + w_4 \cdot \text{age} + b$$

$$= \sum_{i=1}^{n} w_i x_i + b$$

$+ \; b$

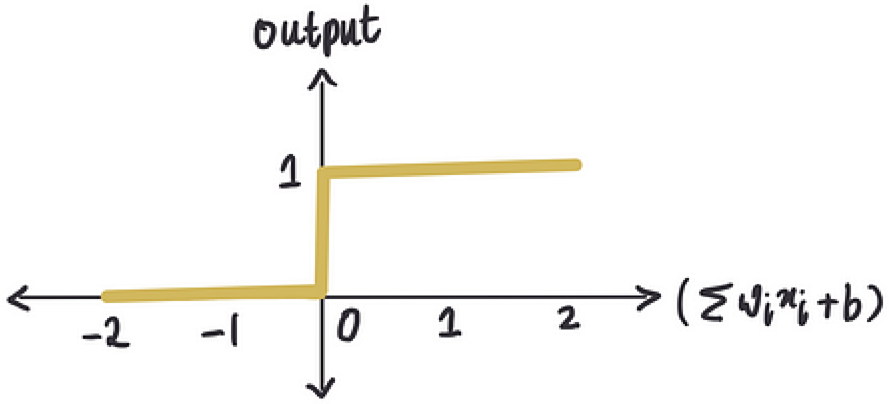
magic, part 2

$$f\Big(\underbrace{\sum_{i=0}^{n} w_i x_i + b}_{\text{input}}\Big) = \text{output}$$

binary step

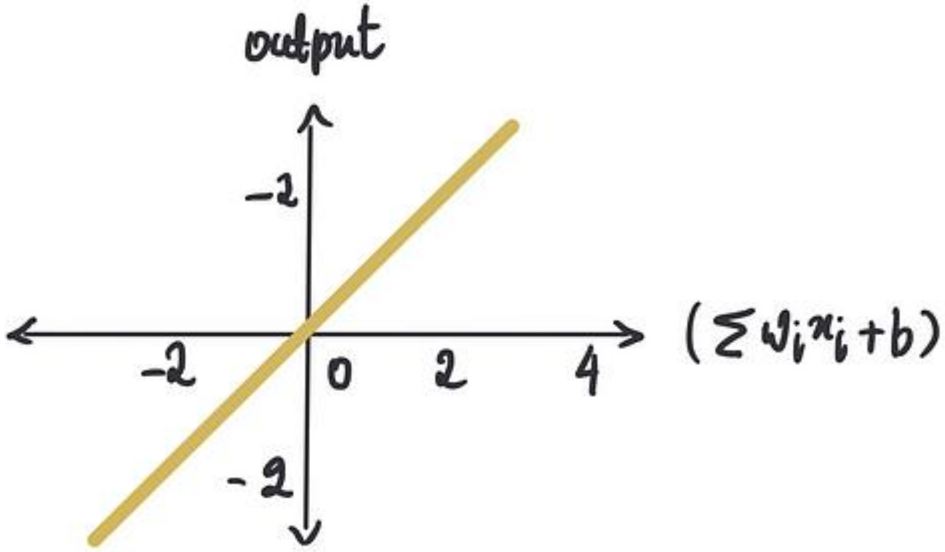
$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

some input



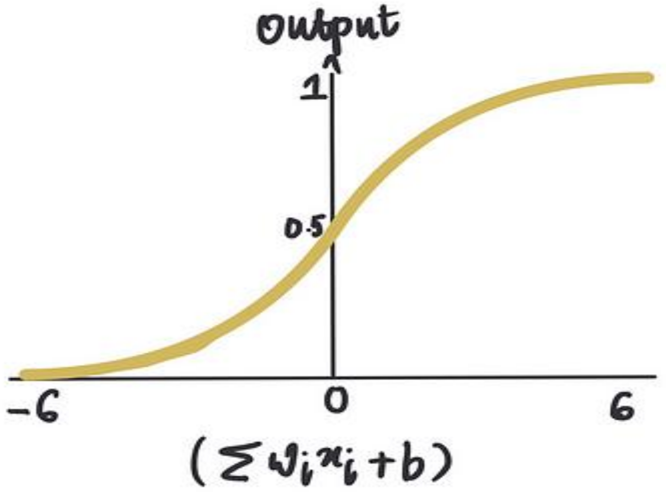
linear activation

$$f(x) = x$$



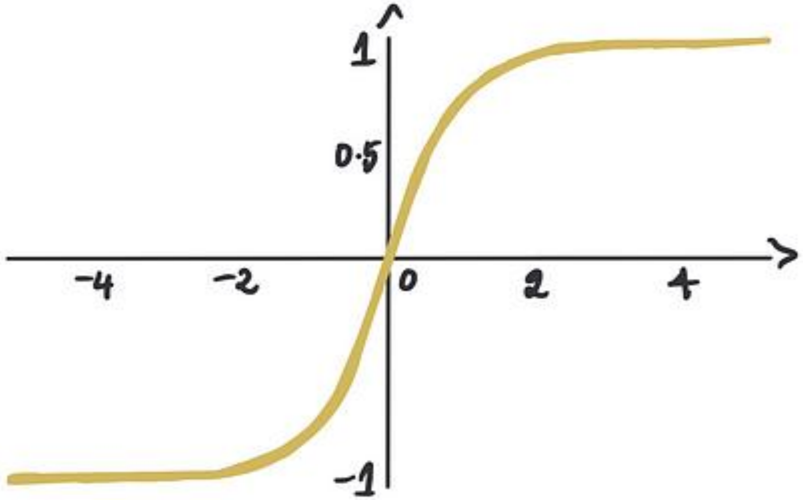
Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$



hyperbolic tangent

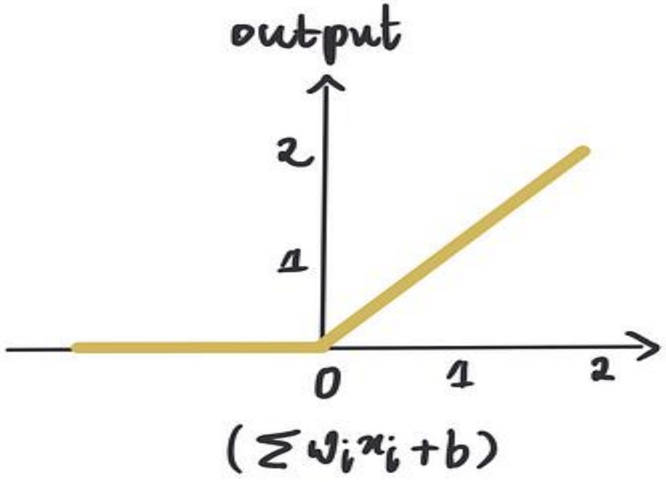
$$f(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$



ReLU

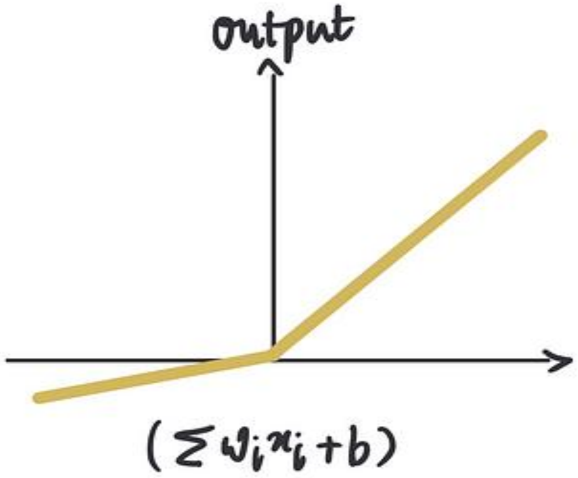
rectified linear units

$$f(x) = \max(x, 0)$$



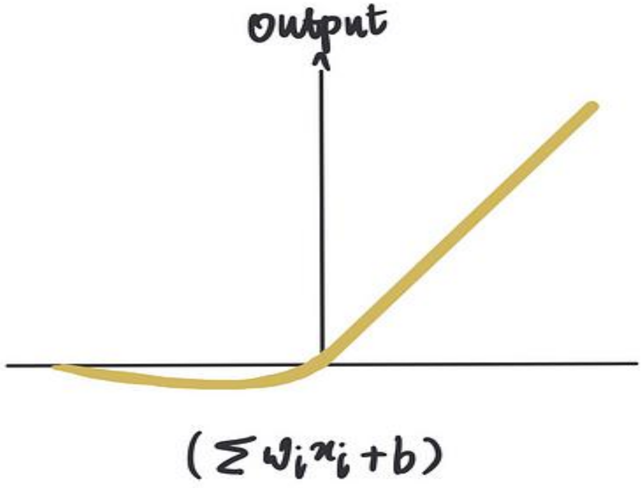
leaky ReLU

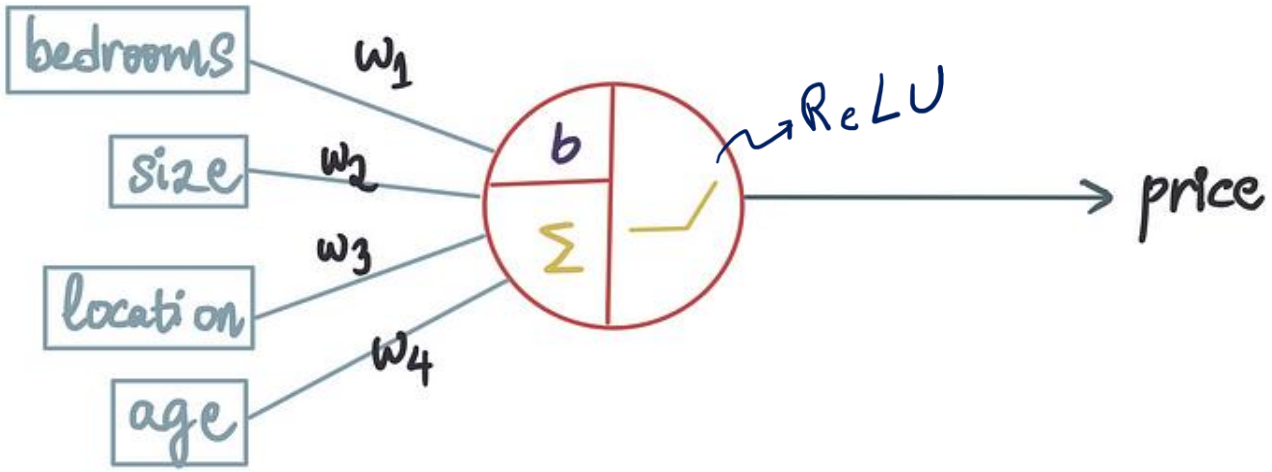
$$f(x) = \begin{cases} 0.01x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$

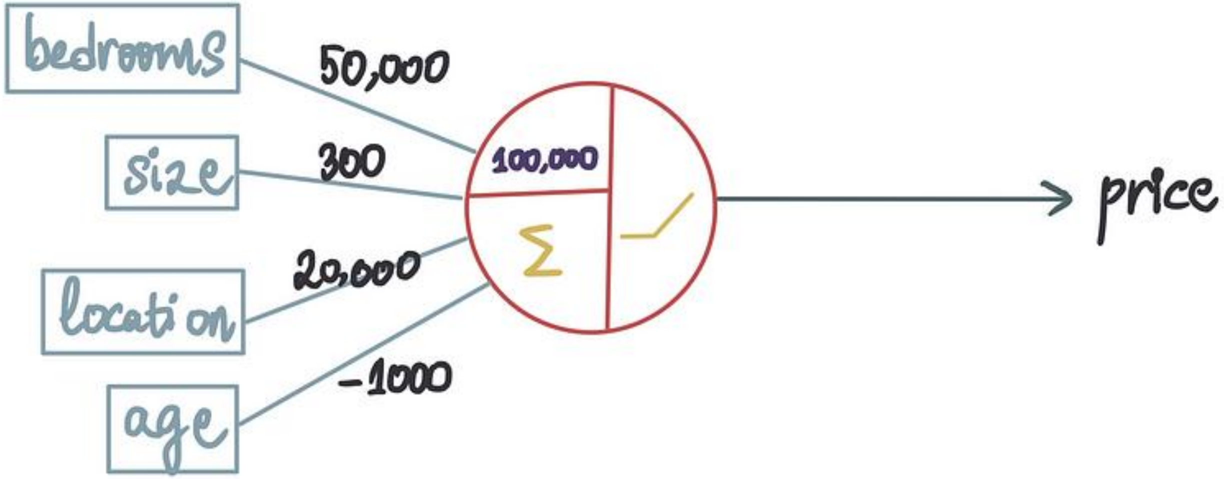


swish

$$f(x) = \frac{x}{1 + e^{-x}}$$

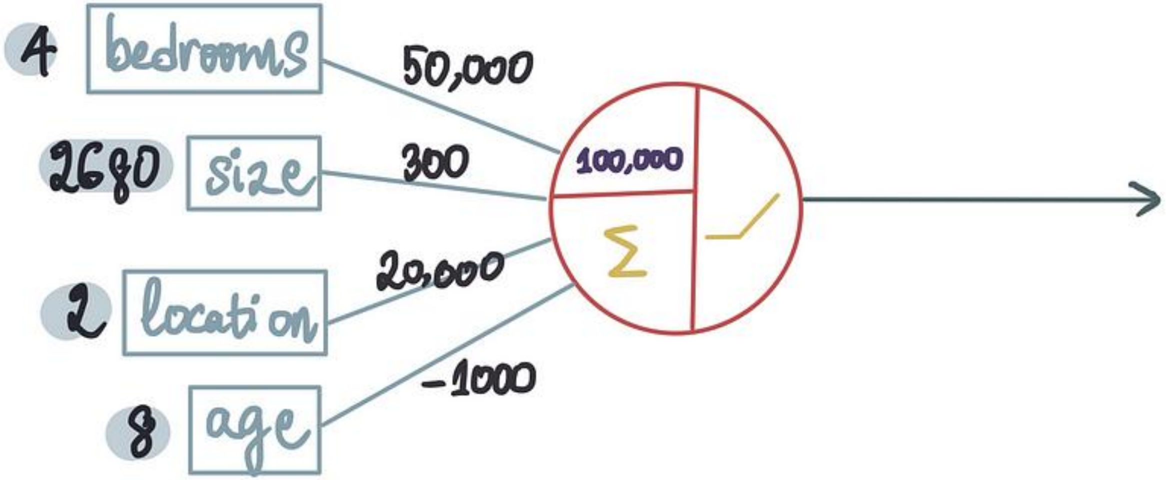



bedrooms
size
location
age
w1
w2
w3
w4
b
Σ
ReLU
price



https://medium.com/

bedrooms	size	location	age	price
4	2680	2	8	1.75M



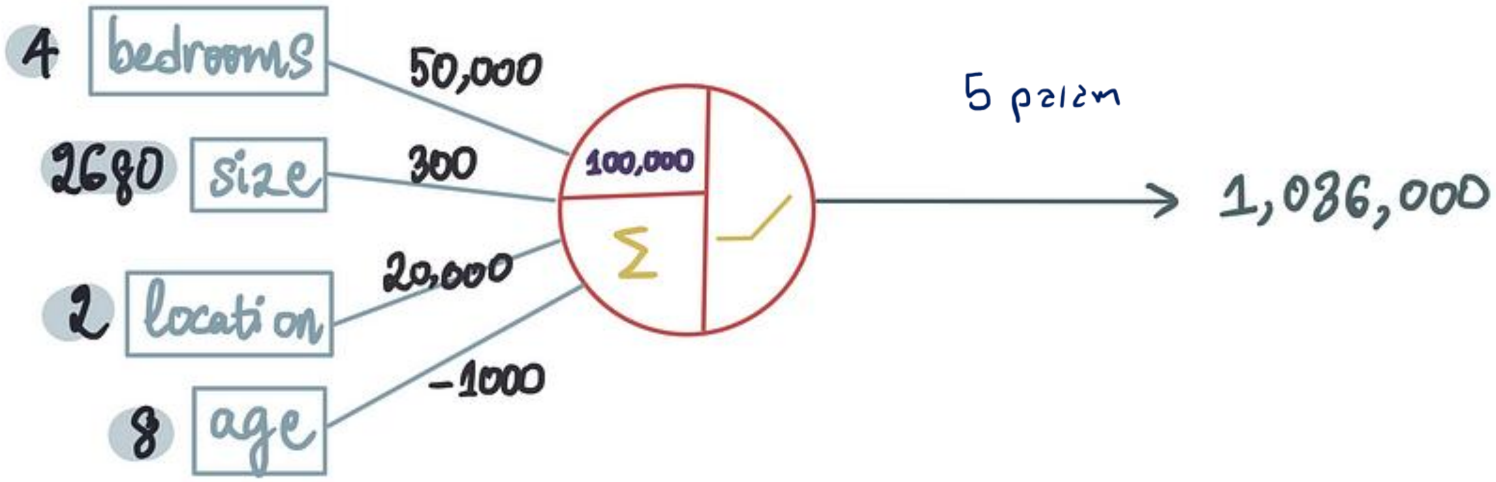
$$\sum_{i=0} w_i x_i + b$$

$$= 50{,}000 \cdot 4 + 300 \cdot 2680 + 2 \cdot 20{,}000 + 8 \cdot -1000 + 100{,}000$$

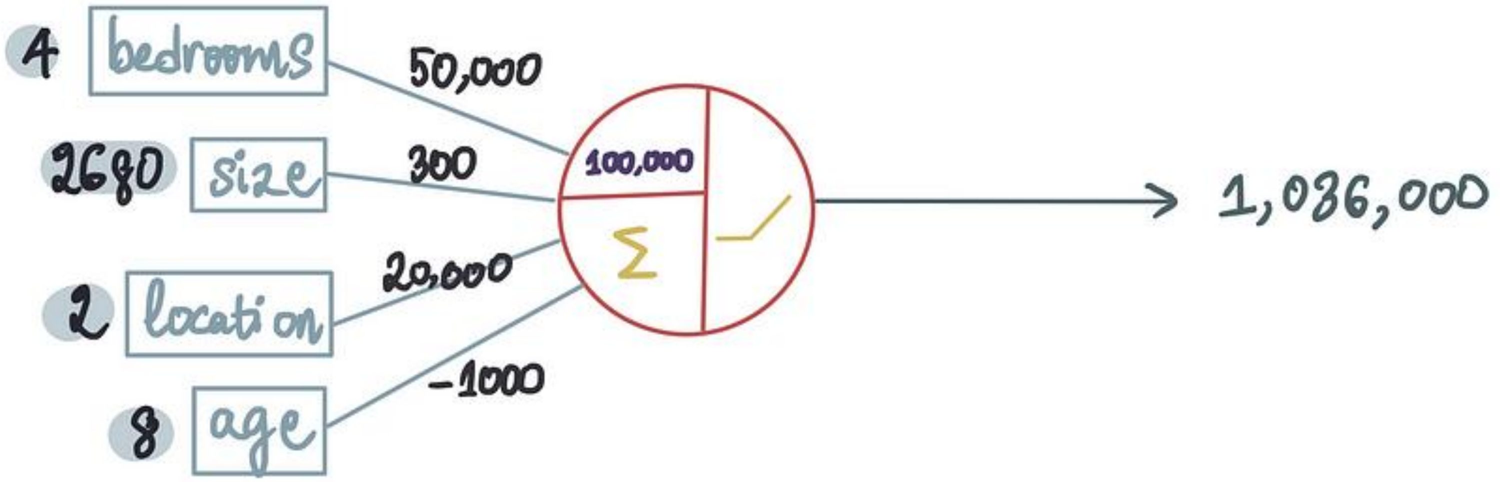
$$= 1{,}036{,}000$$

$$f(x) = \max(x, 0)$$

$$\Rightarrow f(1{,}036{,}000) = 1{,}036{,}000$$

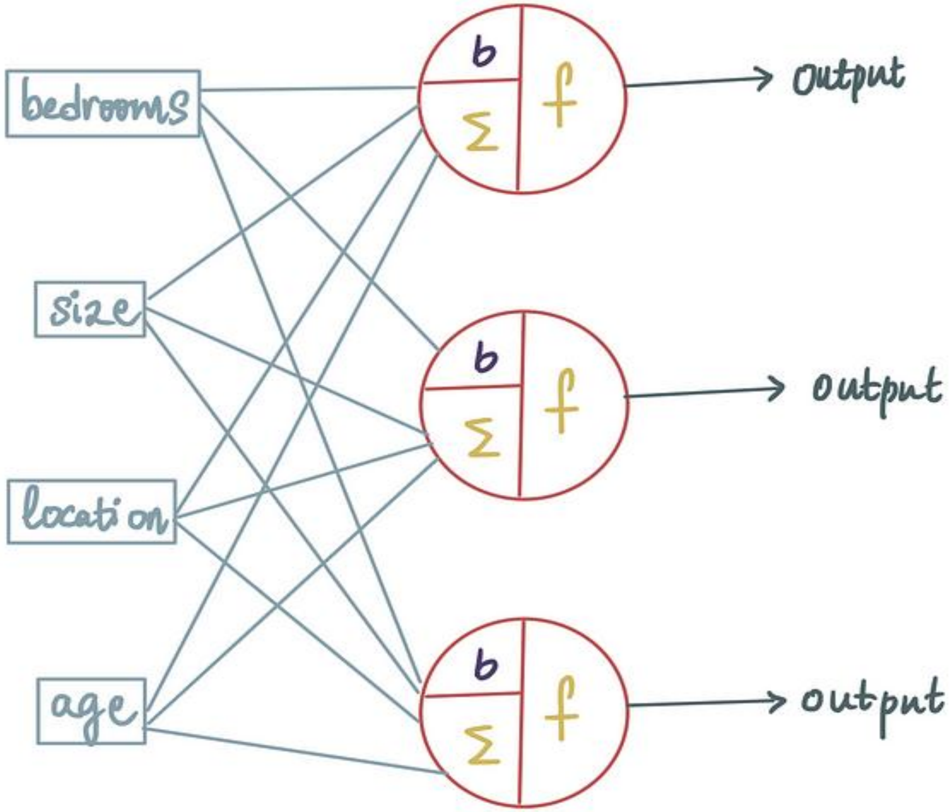


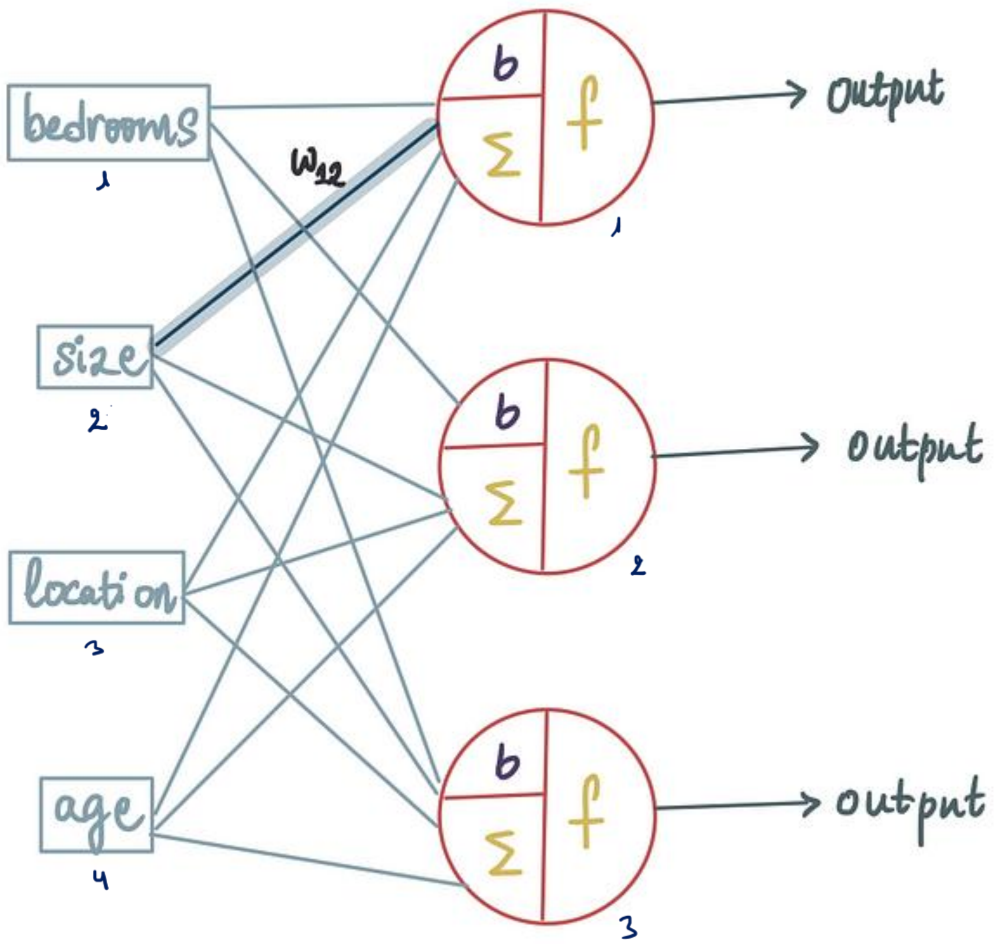
https://medium.com/



$g'(z) - g(z) = \text{Error}$

bedrooms	size	location	age	price		predicted price	
4	2680	2	8	1.75M	−	1,036,000	= Error
5	700	1	27	750K		453,000	
3	2360	1	28	1.2M		870,000	
5	1280	2	98	1.05M		576,000	
5	3480	3	21	2.3M		1,333,000	


bedrooms
size
location
age
b
Σ
f
output
output
output



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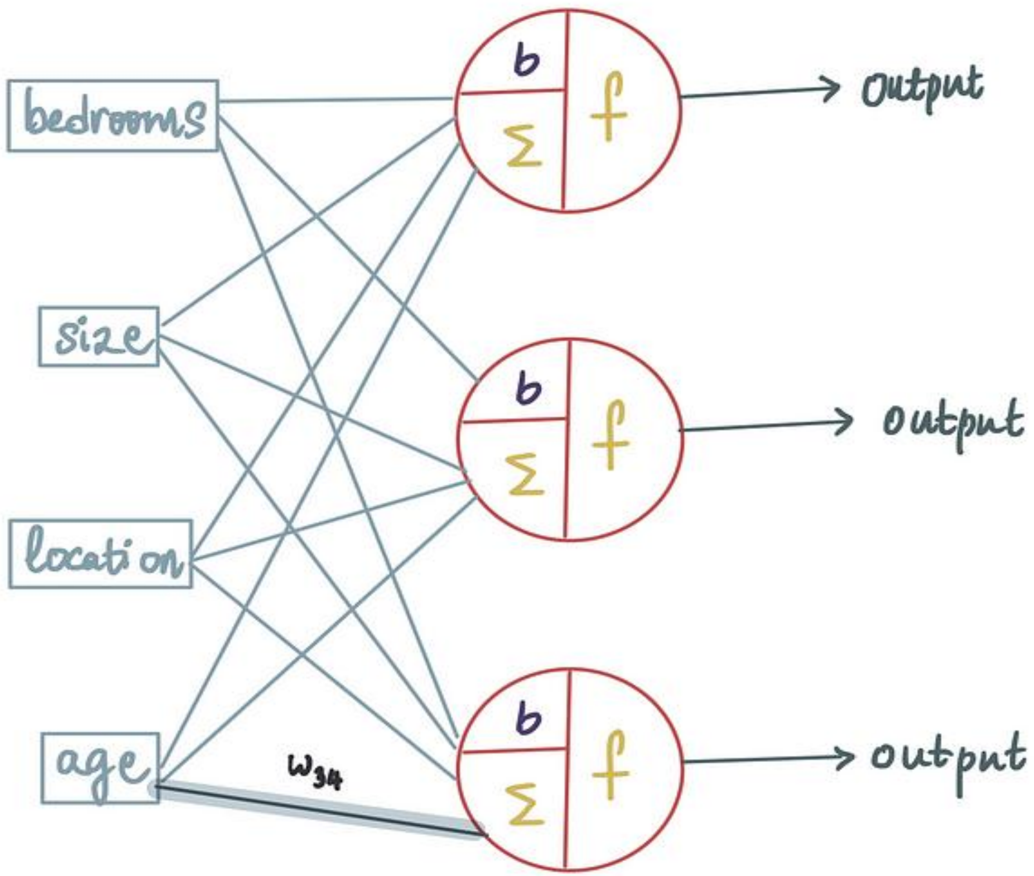

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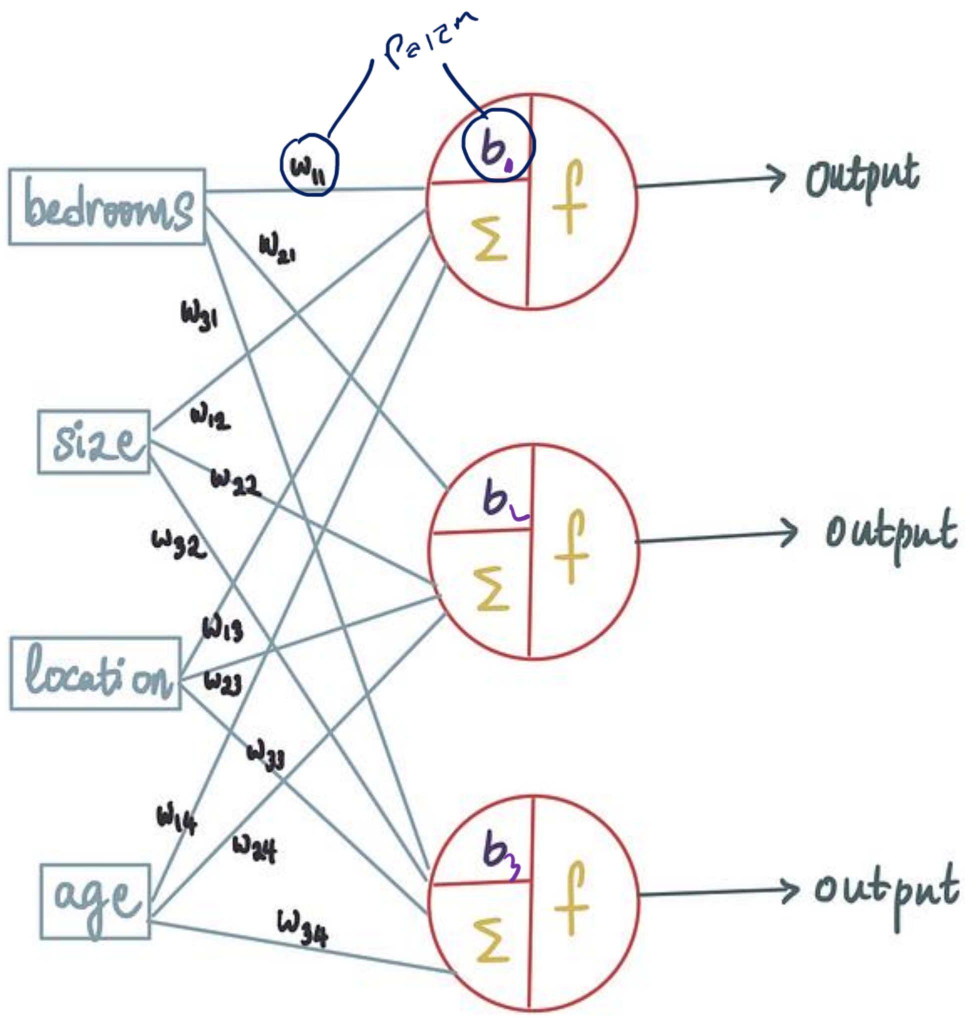

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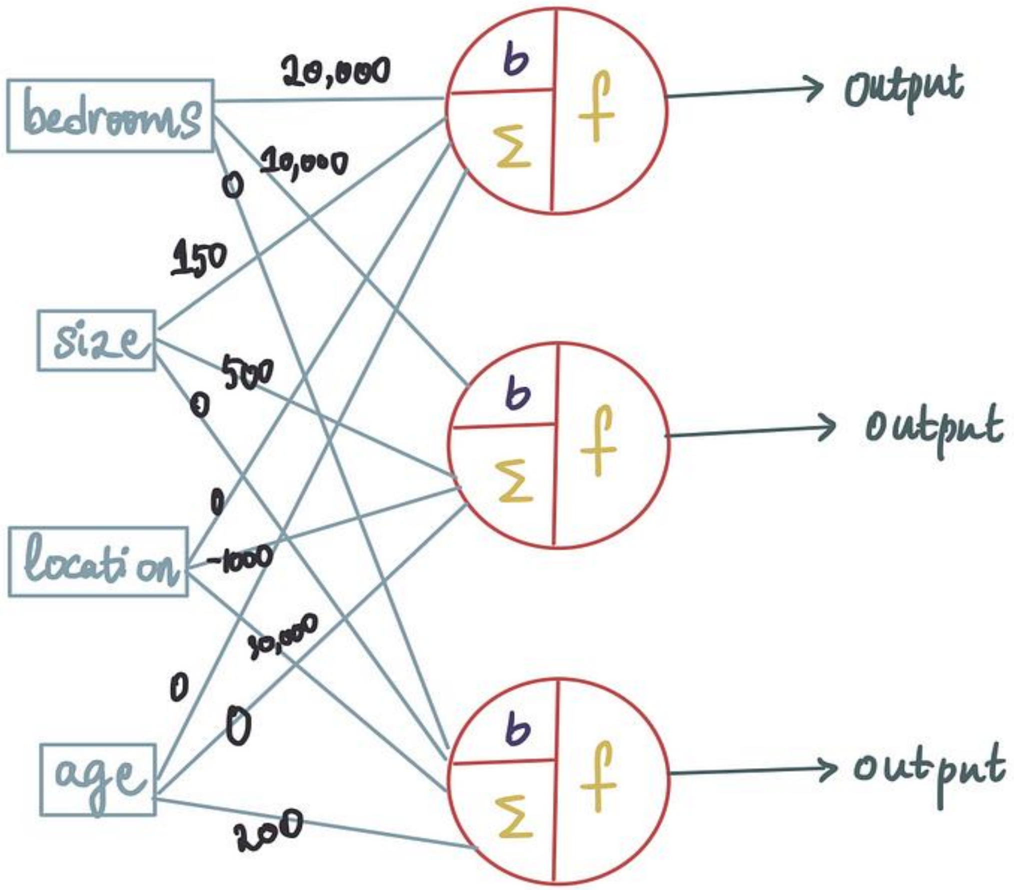

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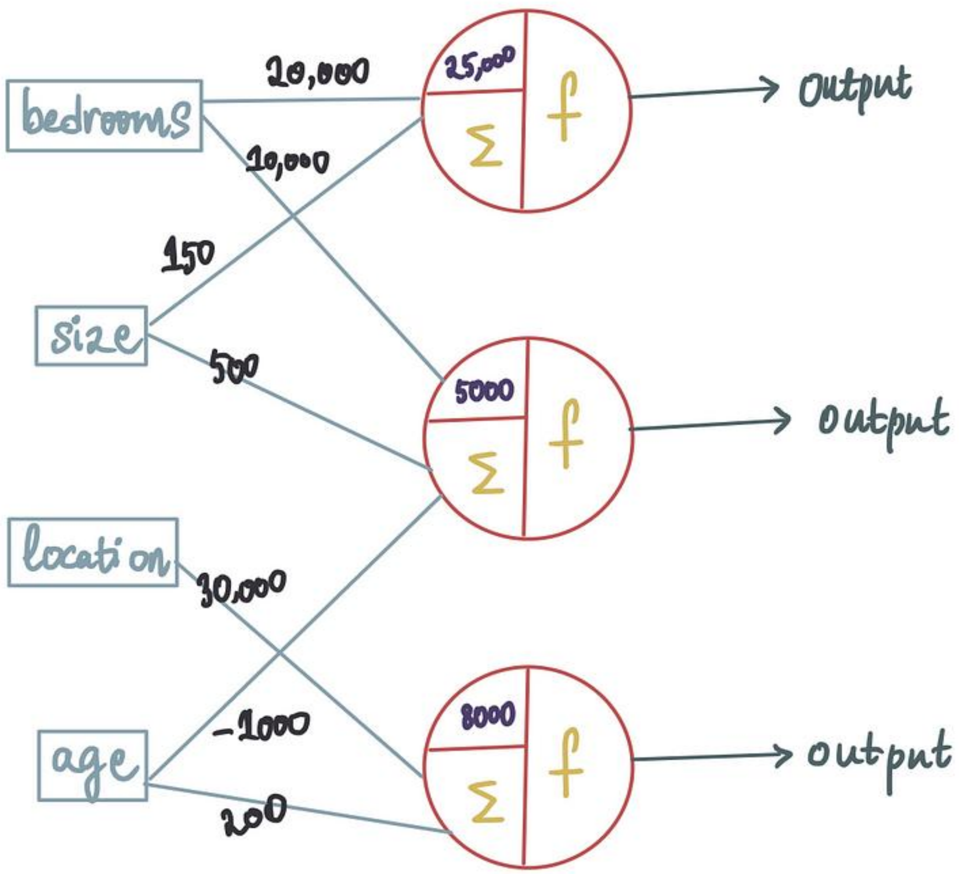

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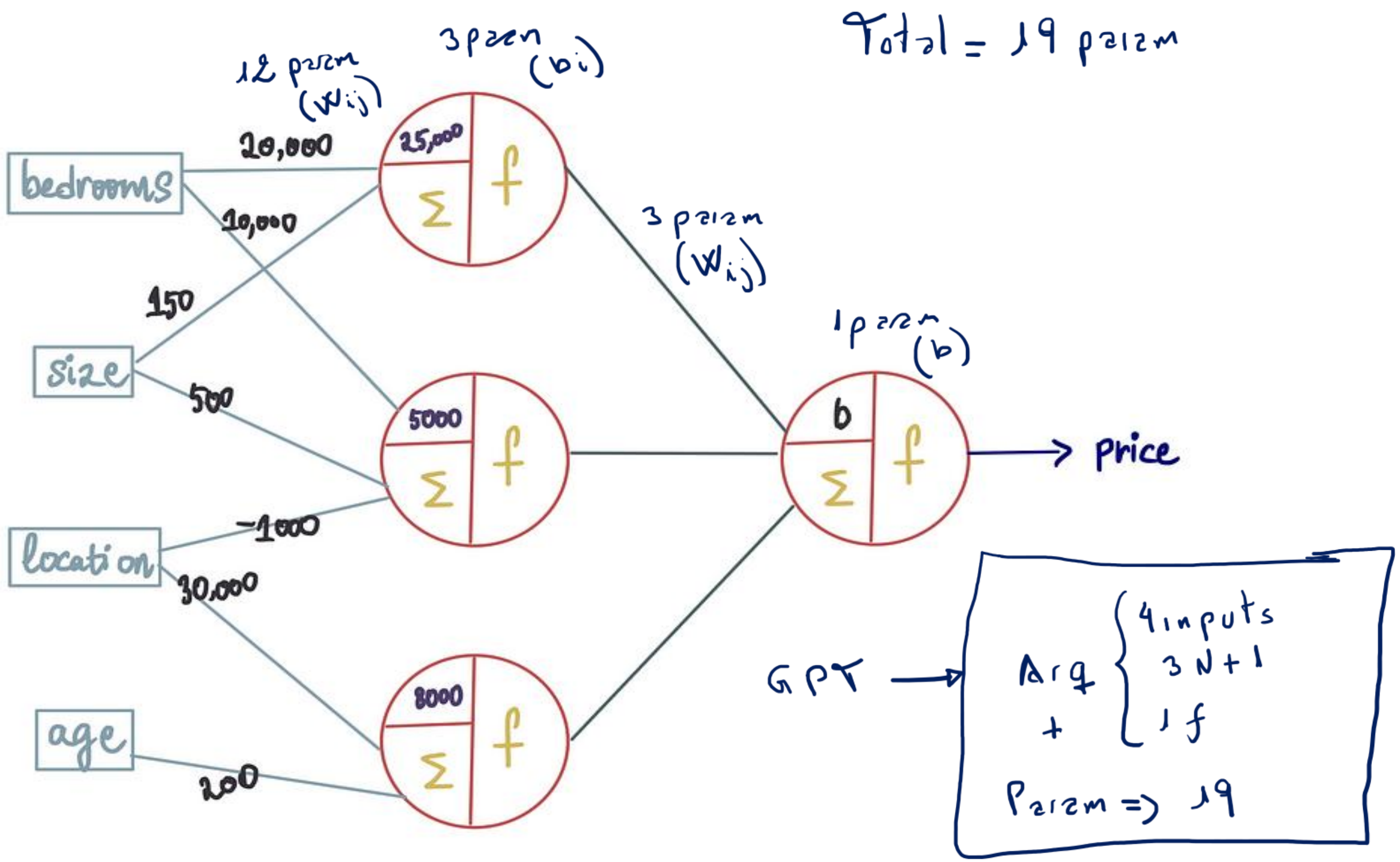

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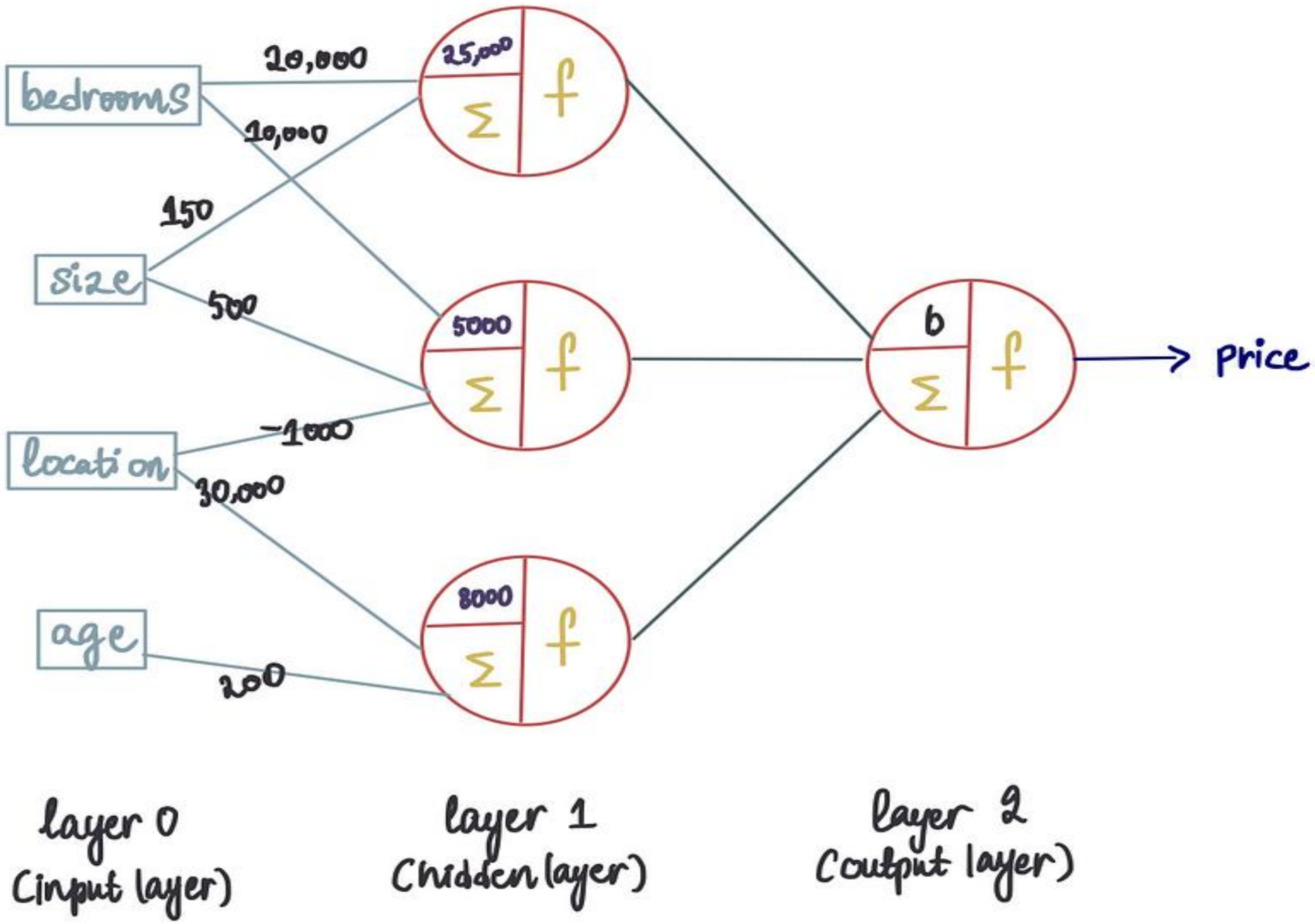

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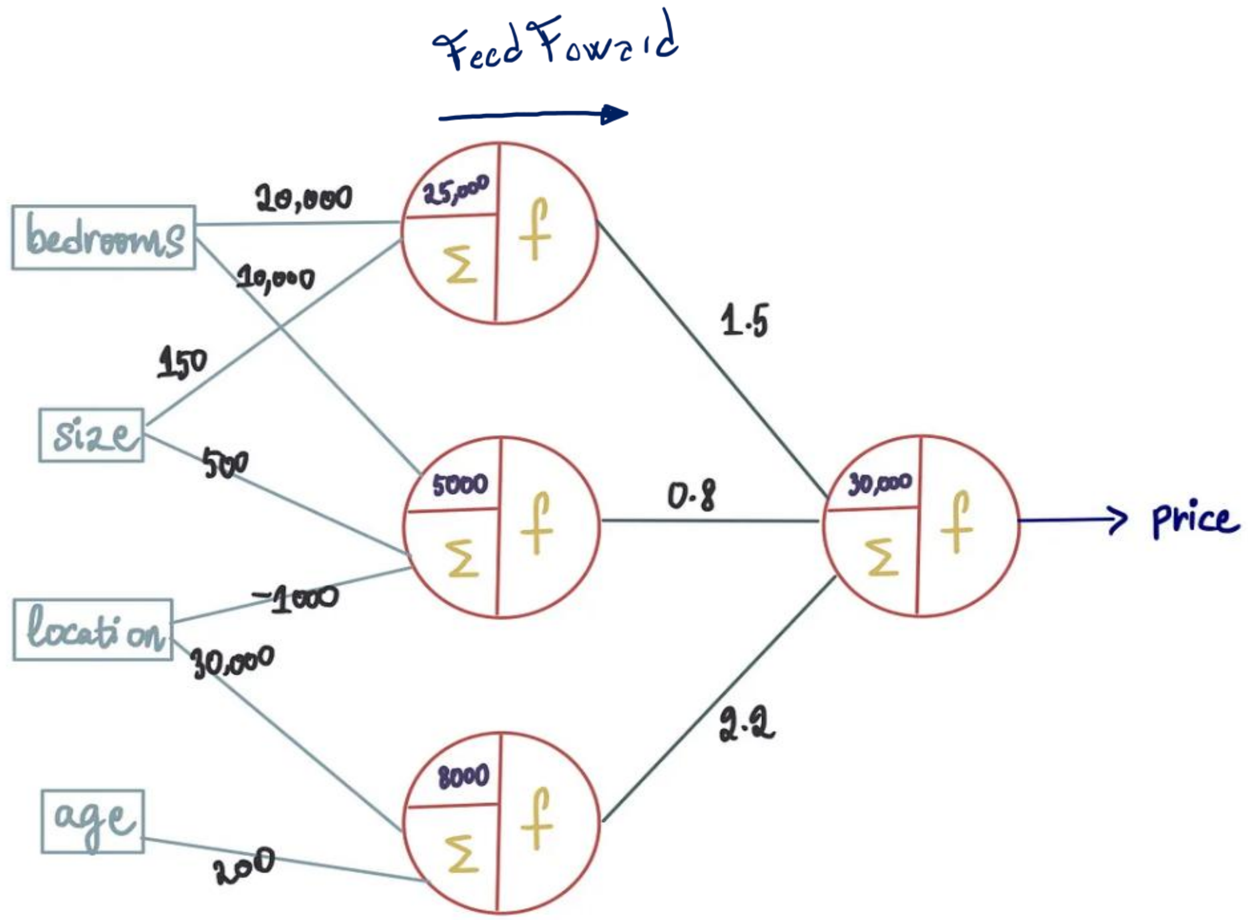
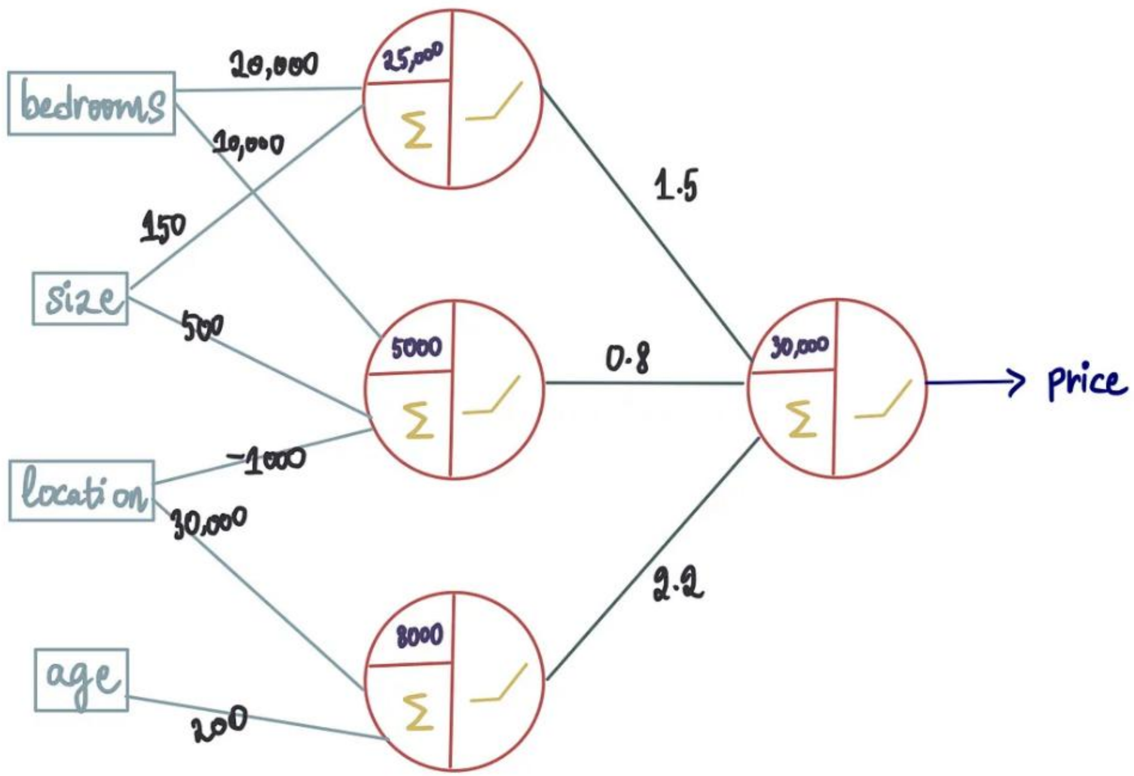
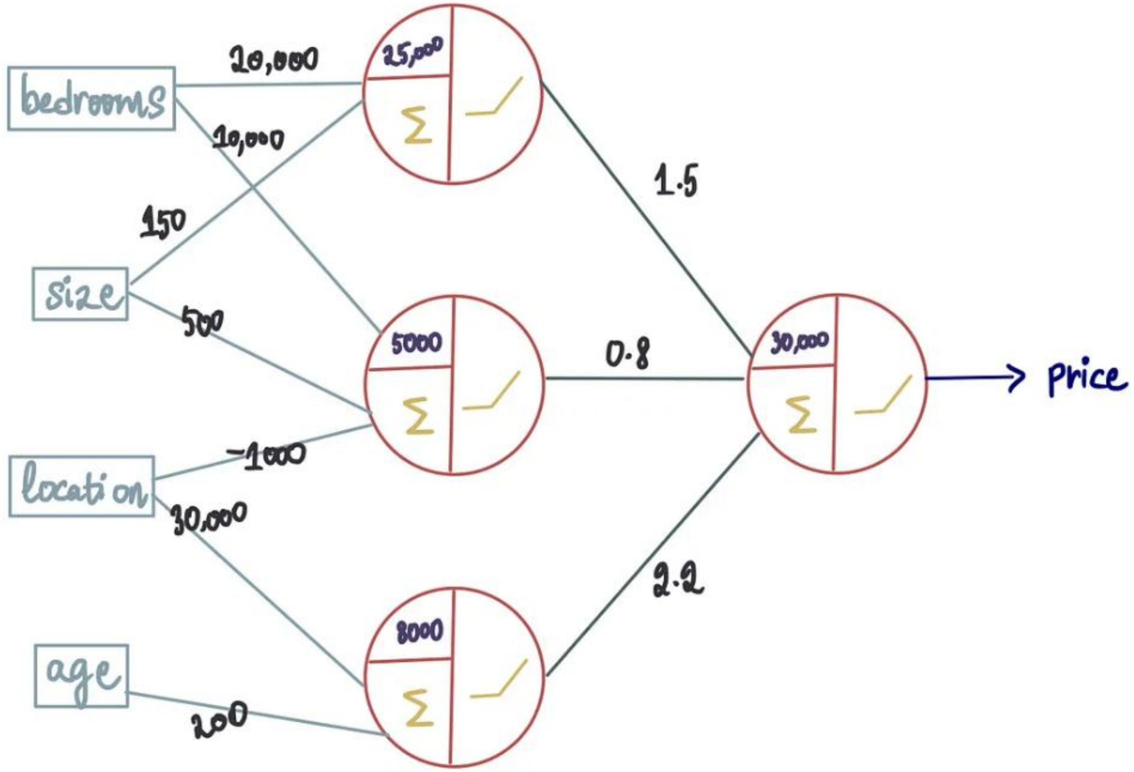
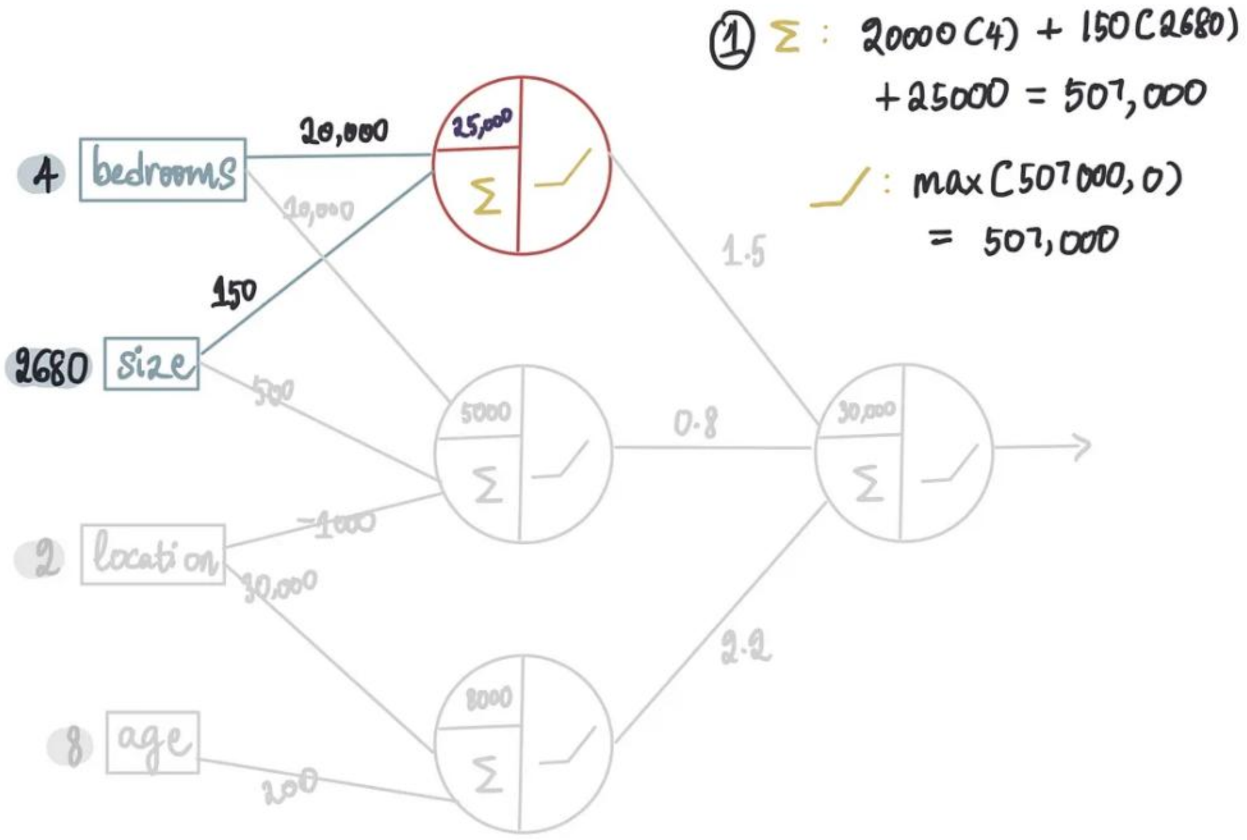

Feed Foward
bedrooms
size
location
age
20,000
10,000
150
500
-1000
30,000
200
25,000
5000
8000
1.5
0.8
2.2
30,000
price



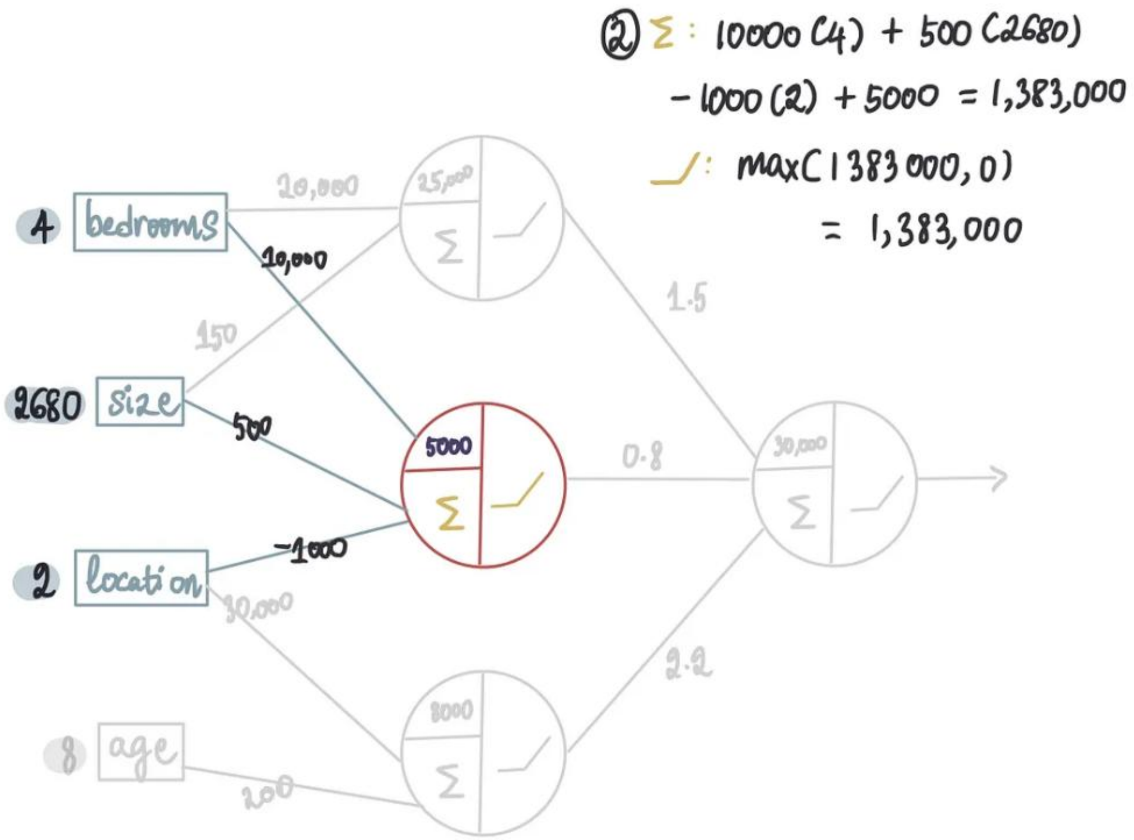
bedrooms
size
location
age
20,000
10,000
150
500
-1000
30,000
200
25,000
5000
8000
1.5
0.8
2.2
30,000
price



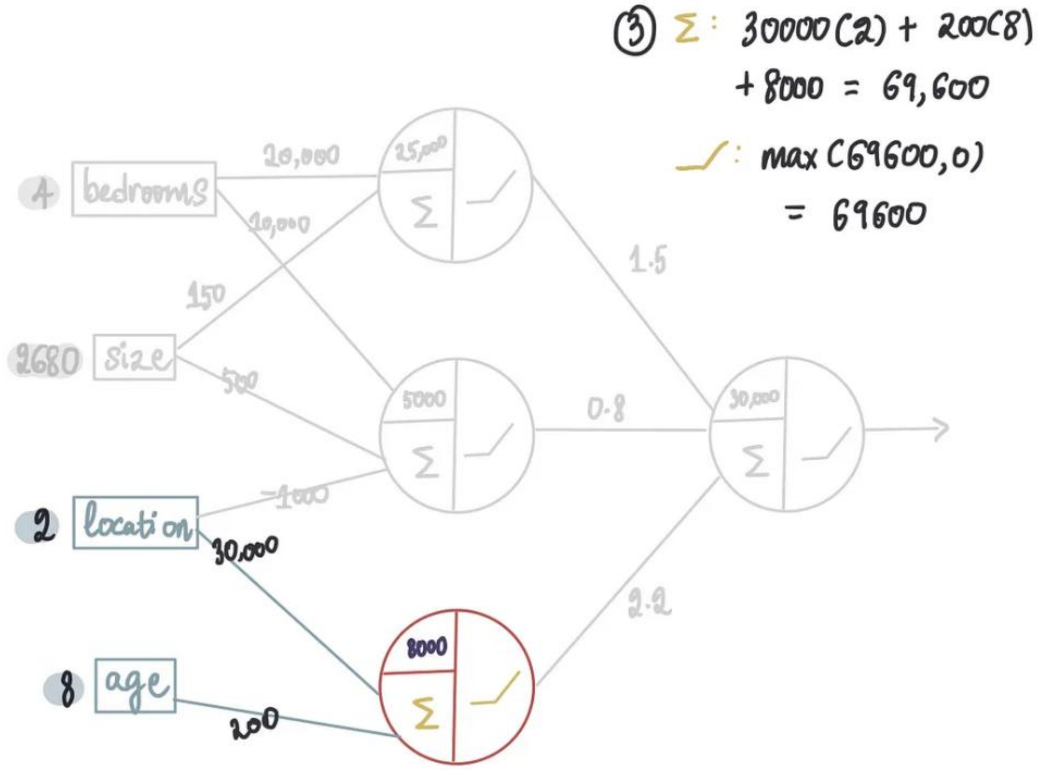
bedrooms	size	location	age	price
4	2680	2	8	1.75M



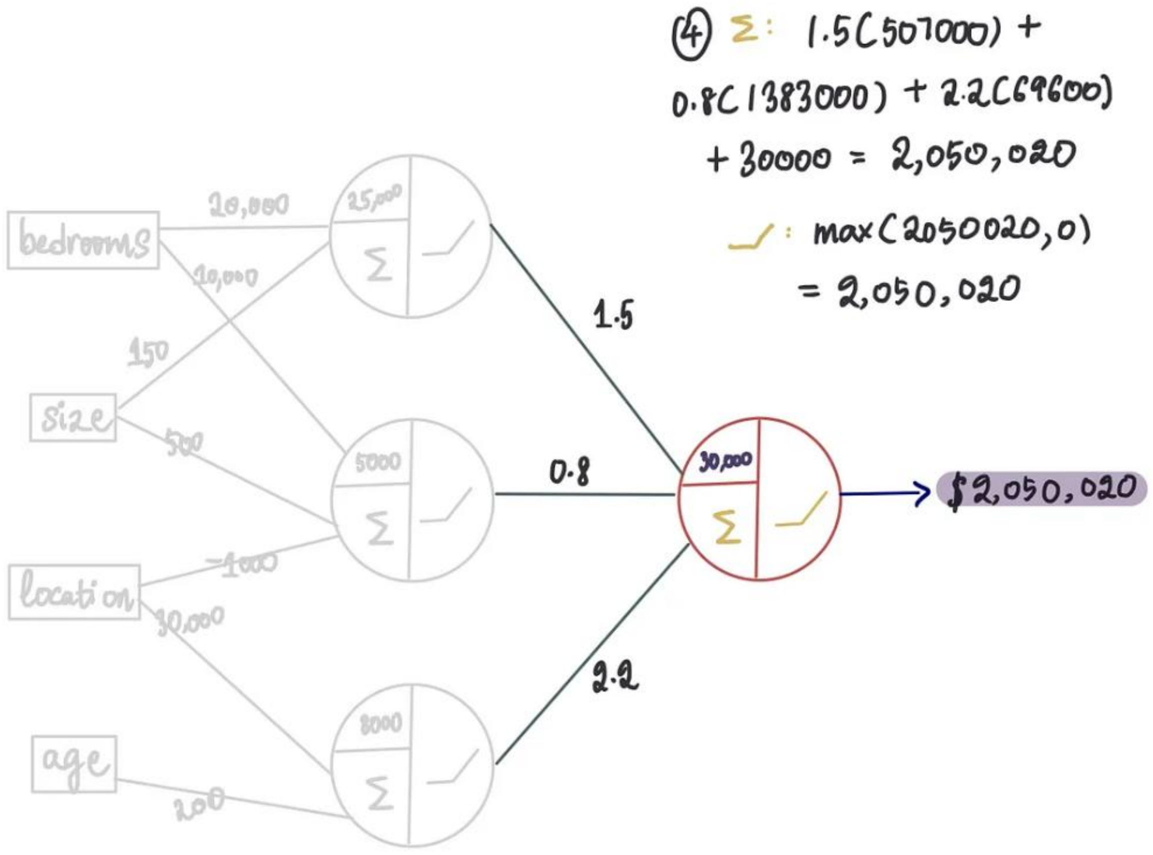
step 1 — first neuron



step 2 — second neuron



step 3 — third neuron



step 4 — final neuron

bedrooms	size	location	age	price	predicted price
4	2680	2	8	1.75M	2,050,020
5	700	1	27	750K	793,680
3	2360	1	28	1.2M	1,820,820
5	1280	2	98	1.05M	1,252,620
5	3480	3	21	2.3M	2,658,940

actual price	old predicted price	new predicted price
1.75M	1,036,000	2,050,020
750K	453,000	793,680
1.2M	870,000	1,820,820
1.05M	576,000	1,252,620
2.3M	1,333,000	2,658,940

actual value

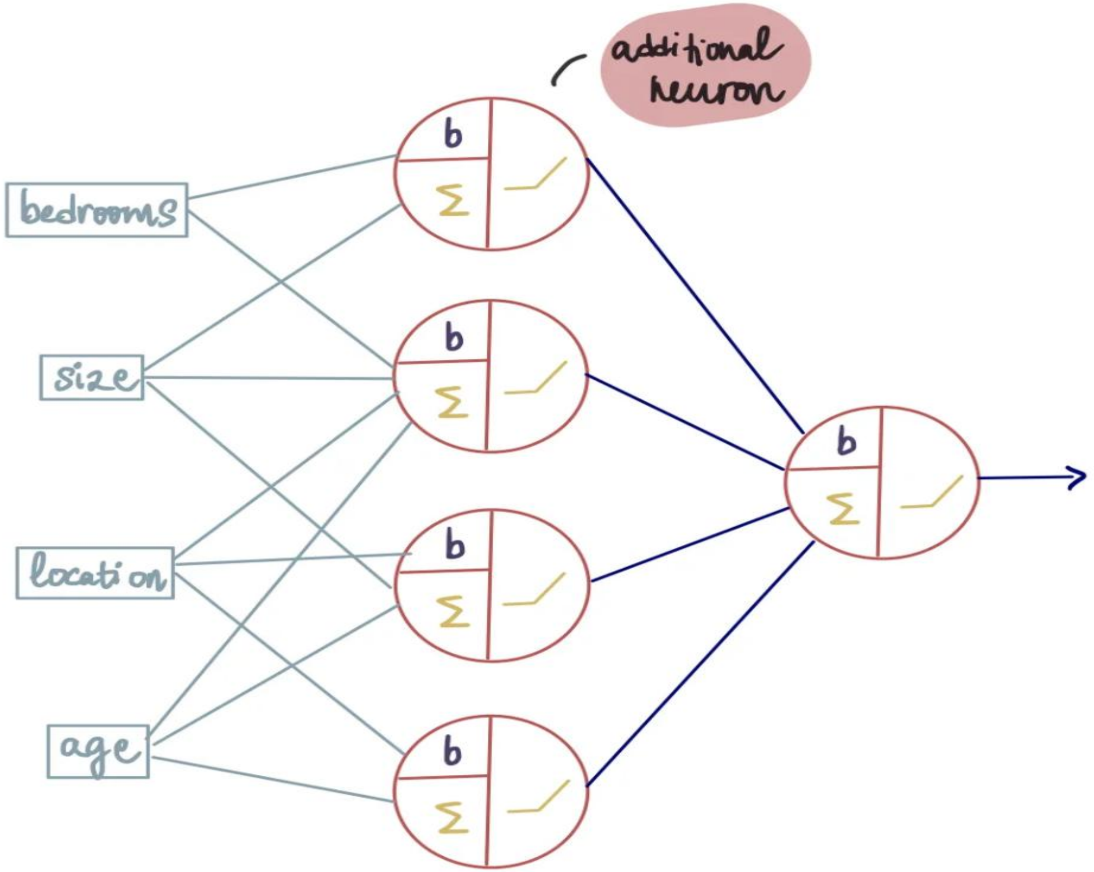
predicted value

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

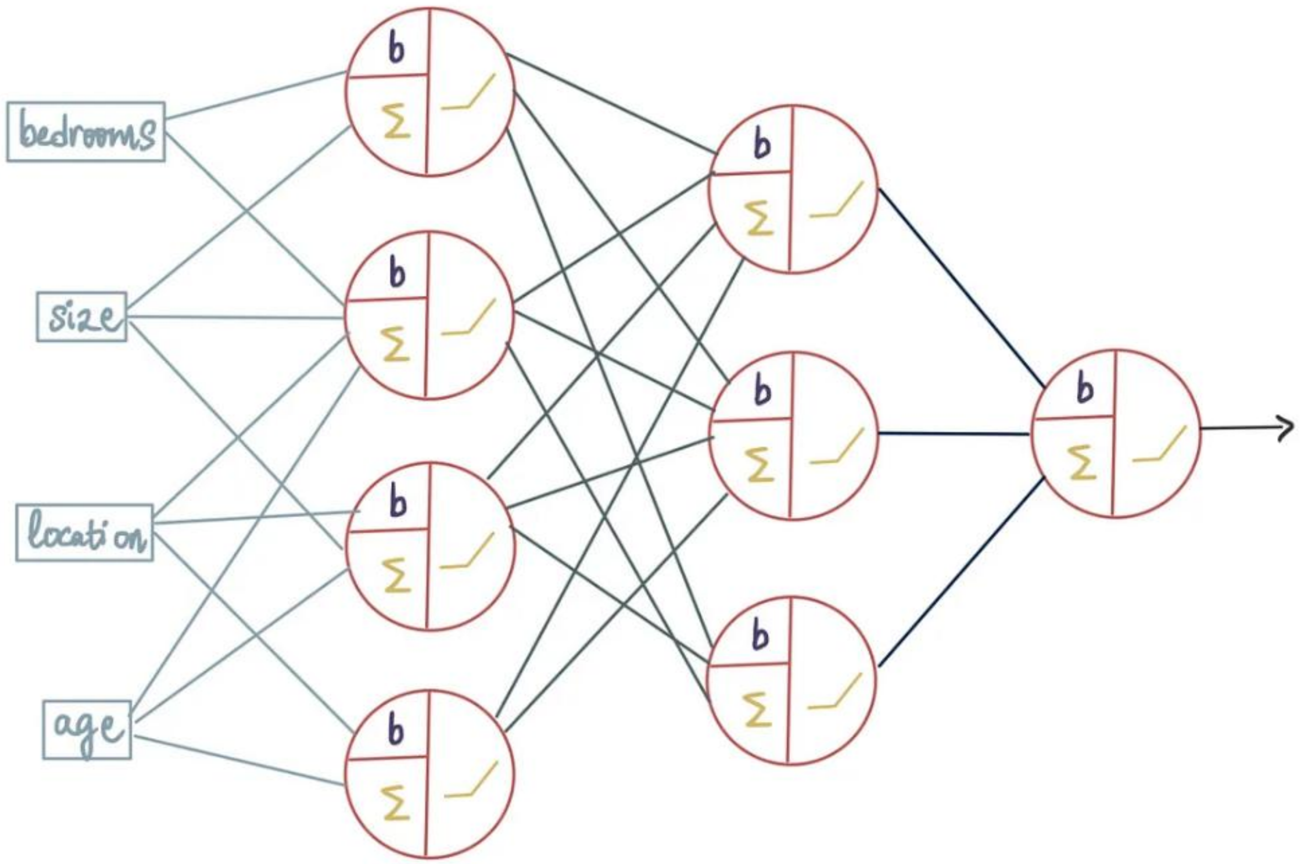
$$MSE_{\text{old model}} = \frac{1}{5} \left[(1.75M - 1{,}036{,}000)^2 + (750K - 453{,}000)^2 + \ldots\ldots + (2.3M - 1{,}333{,}000)^2 \right]$$

$$= 1.867 \times 10^{12}$$

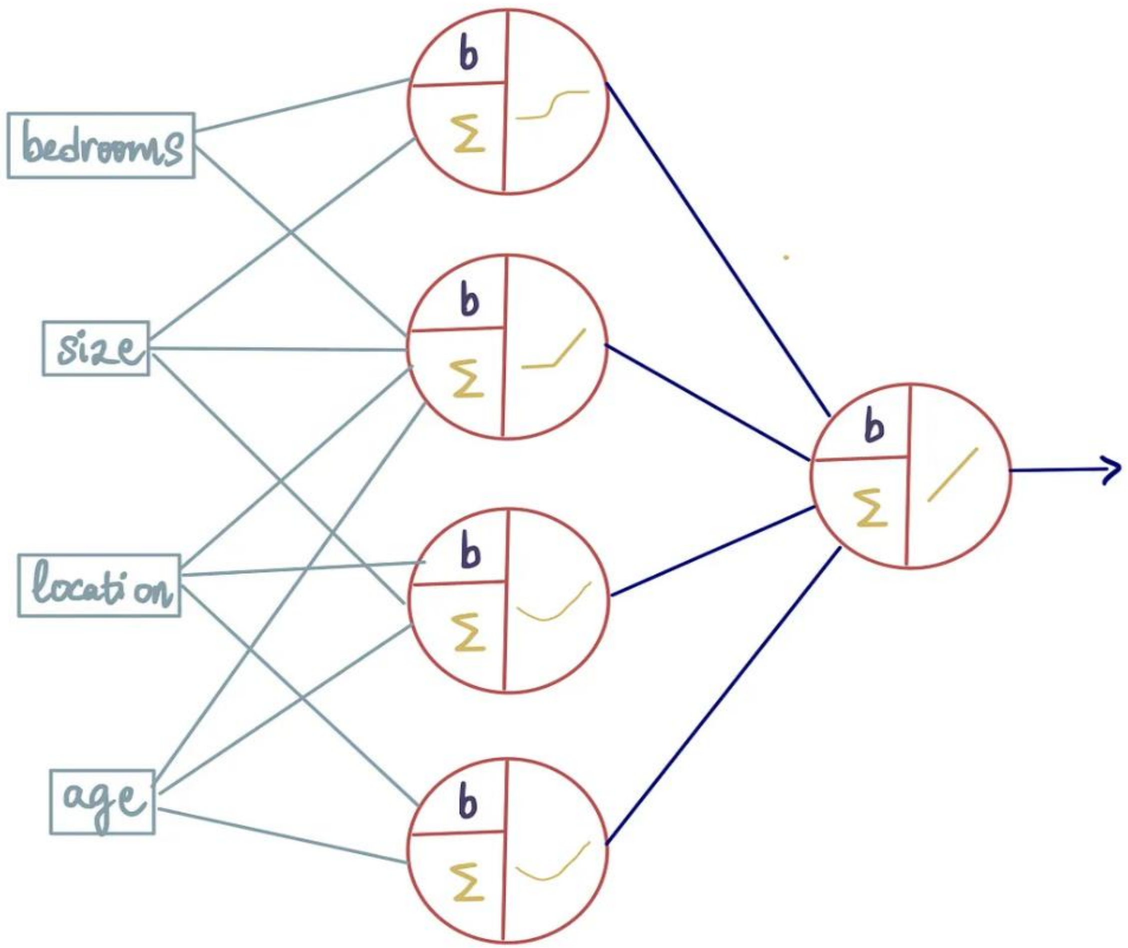
$$MSE_{\text{new model}} = 6.47 \times 10^{11}$$



added a fourth neuron to the hidden layer



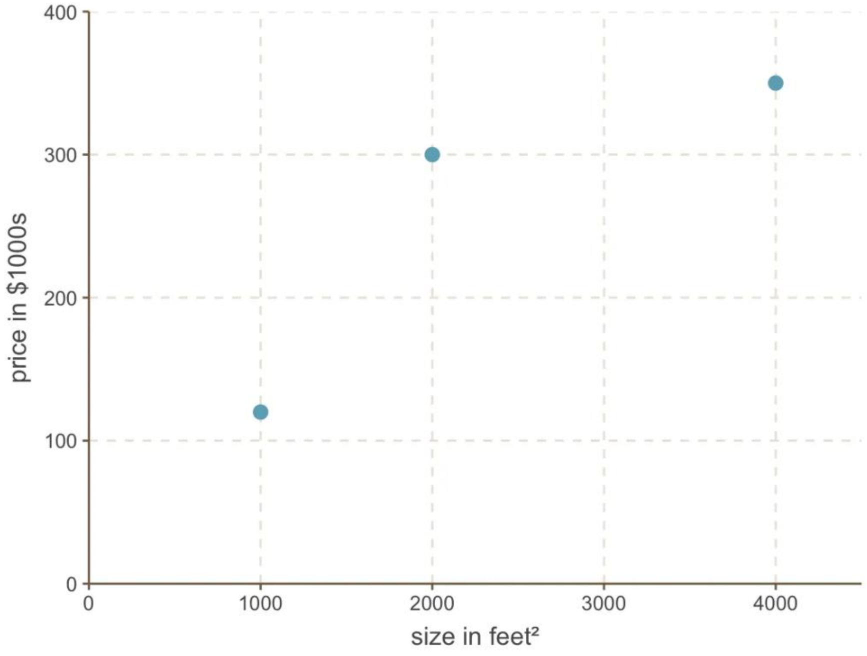
added a second hidden layer with 3 neurons

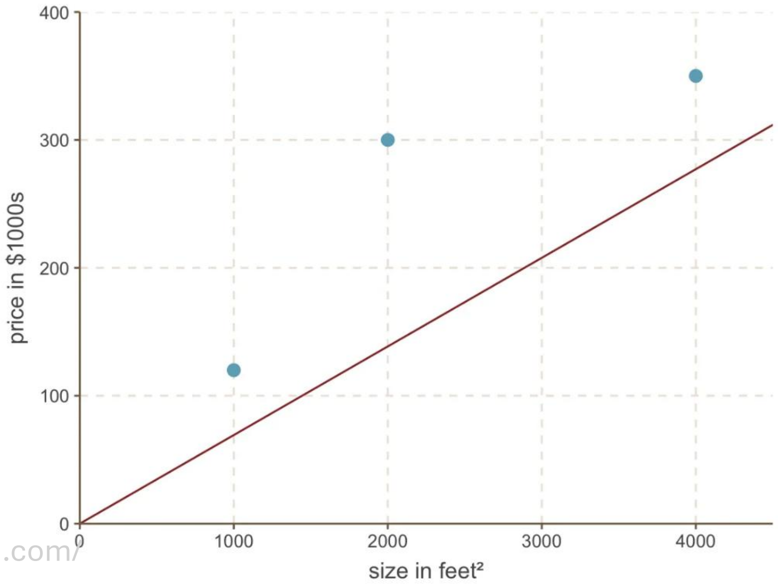

bedrooms
size
location
age
b
Σ
b
Σ
b
Σ
b
Σ
b
Σ

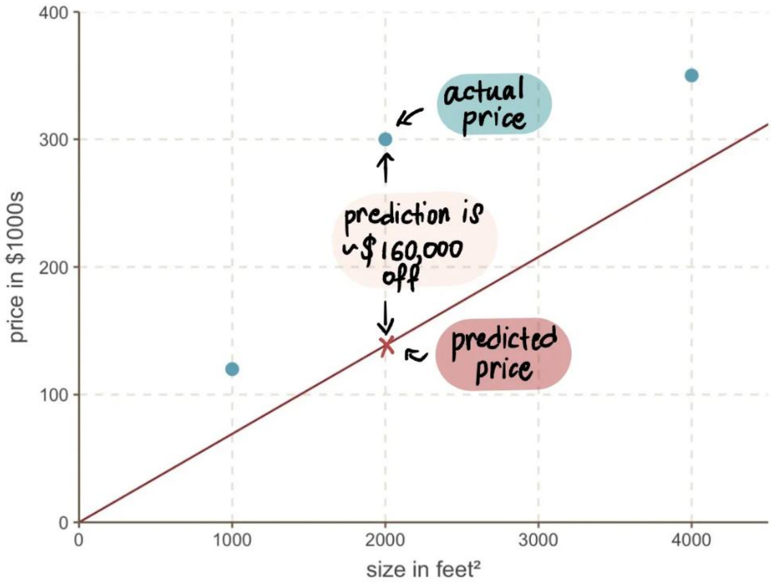
Introducción a Redes Neuronales

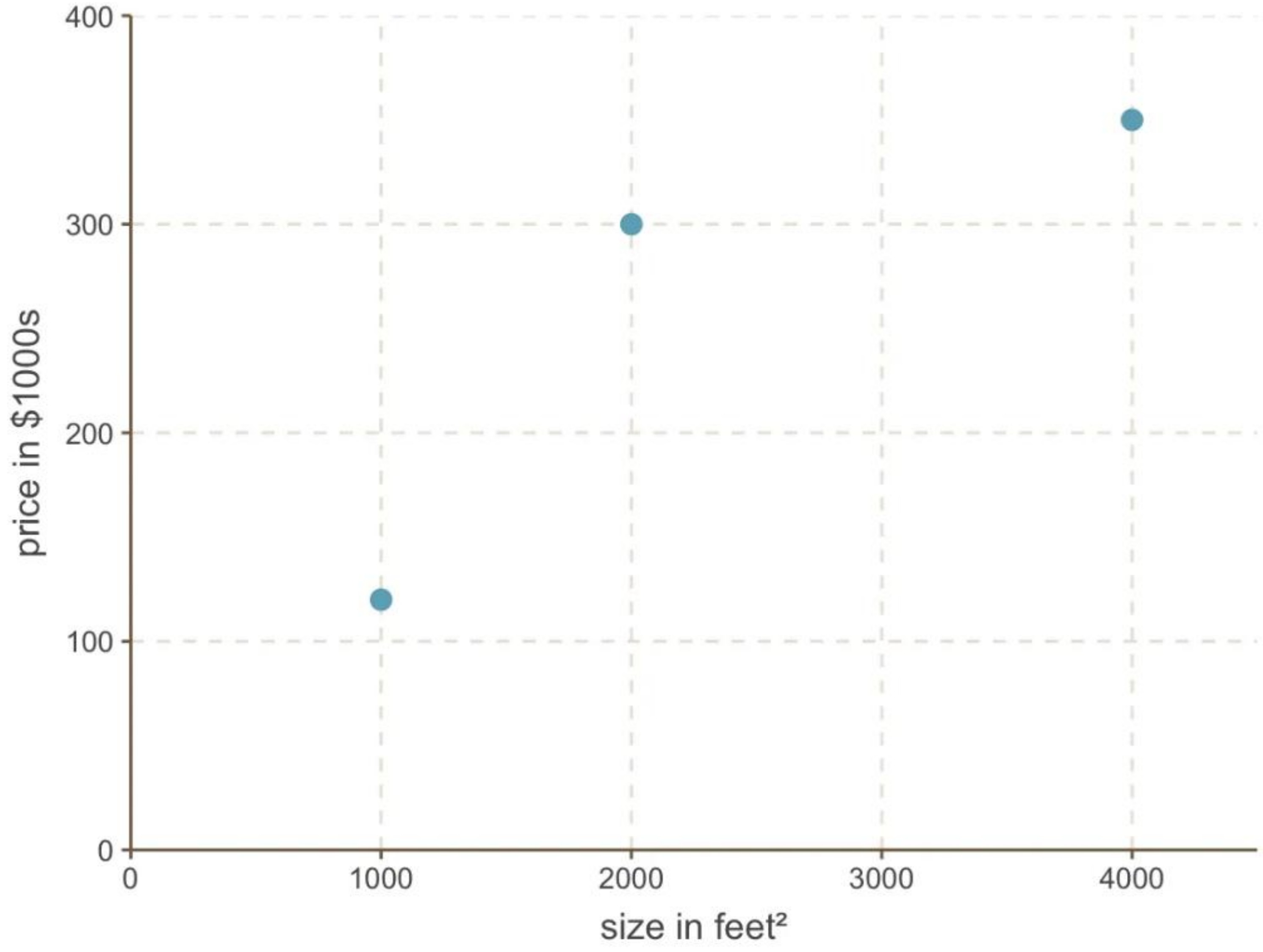
Gradiente Descendiente

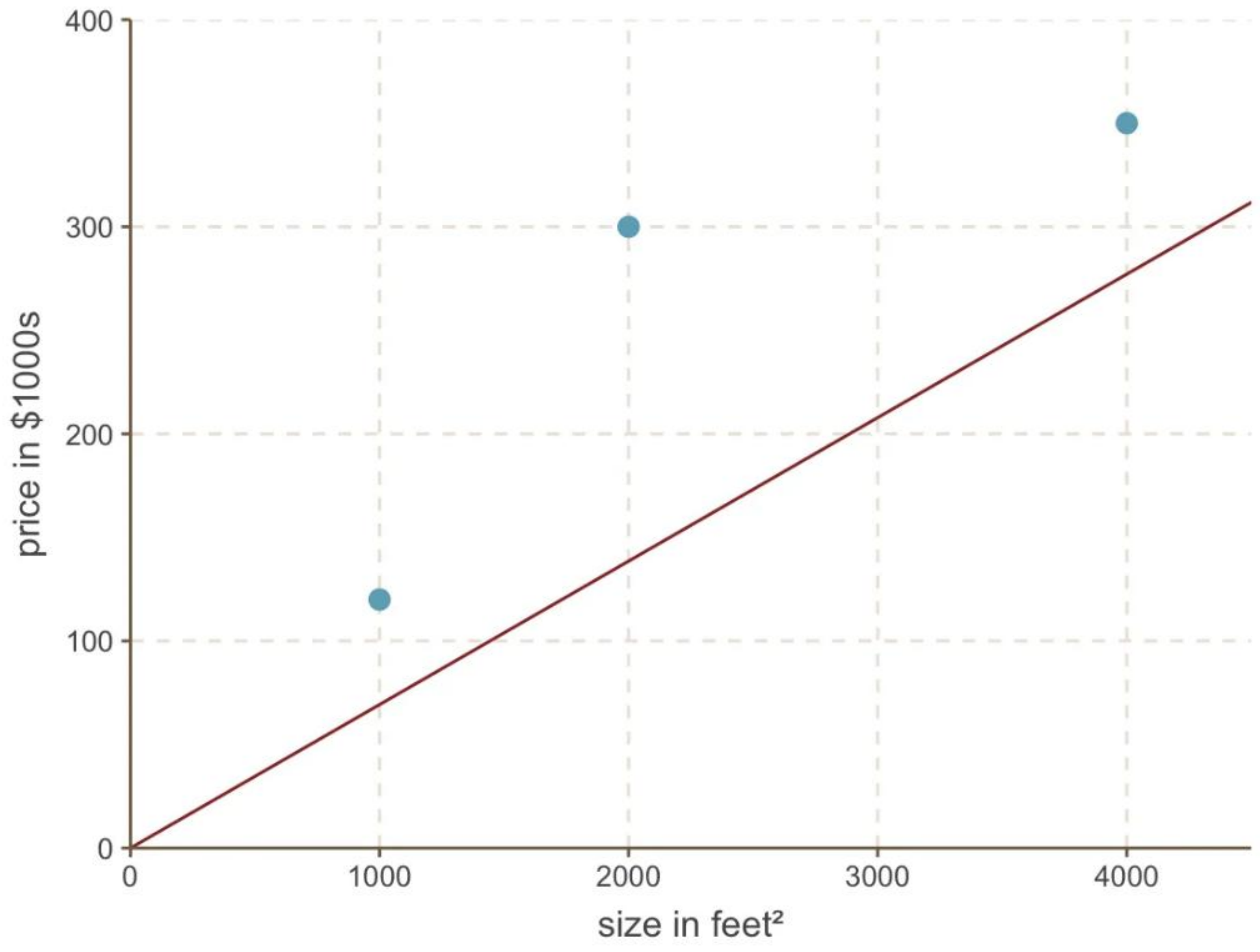
house size	price in $1000s
1000 feet²	120
2000 feet²	300
3000 feet²	350

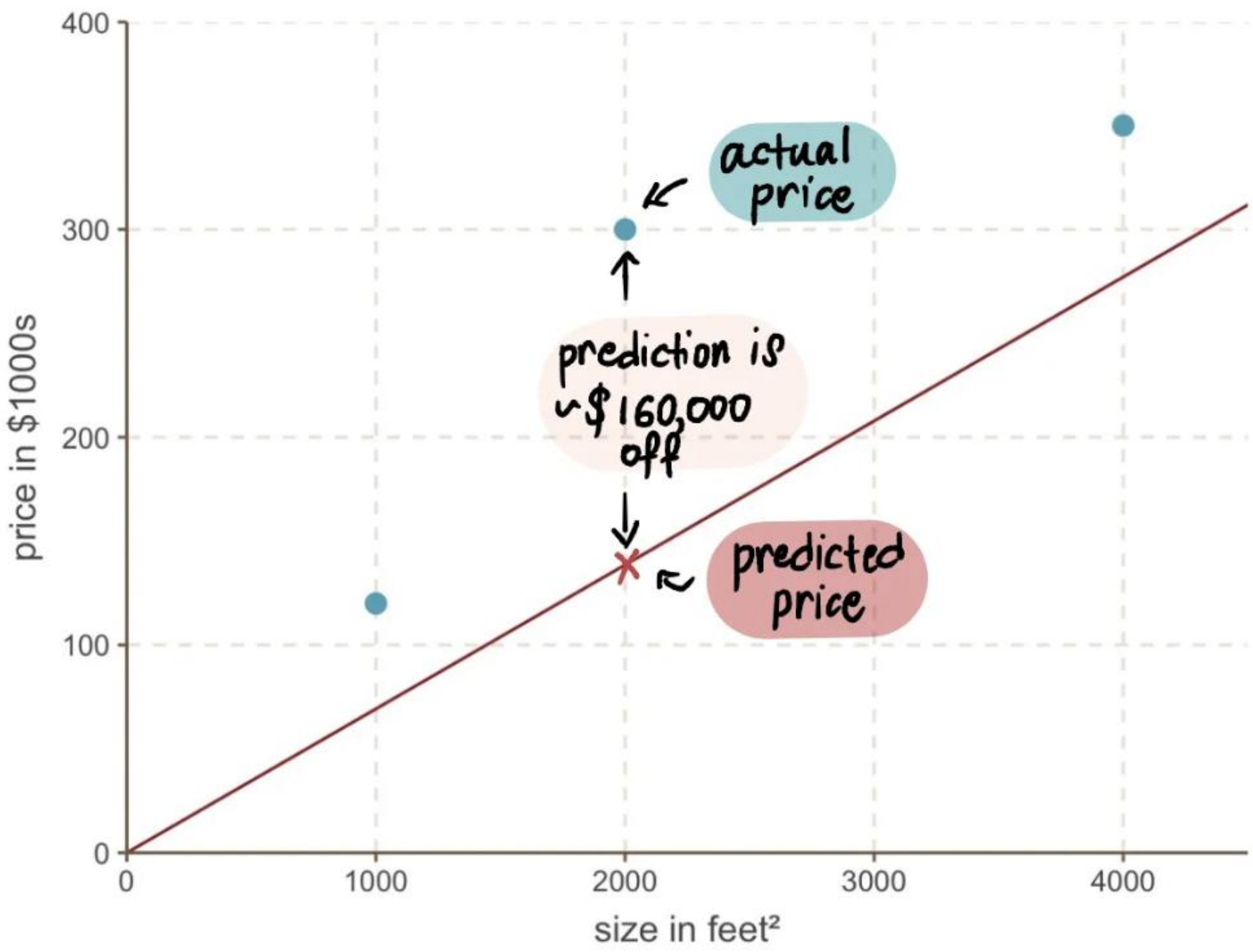


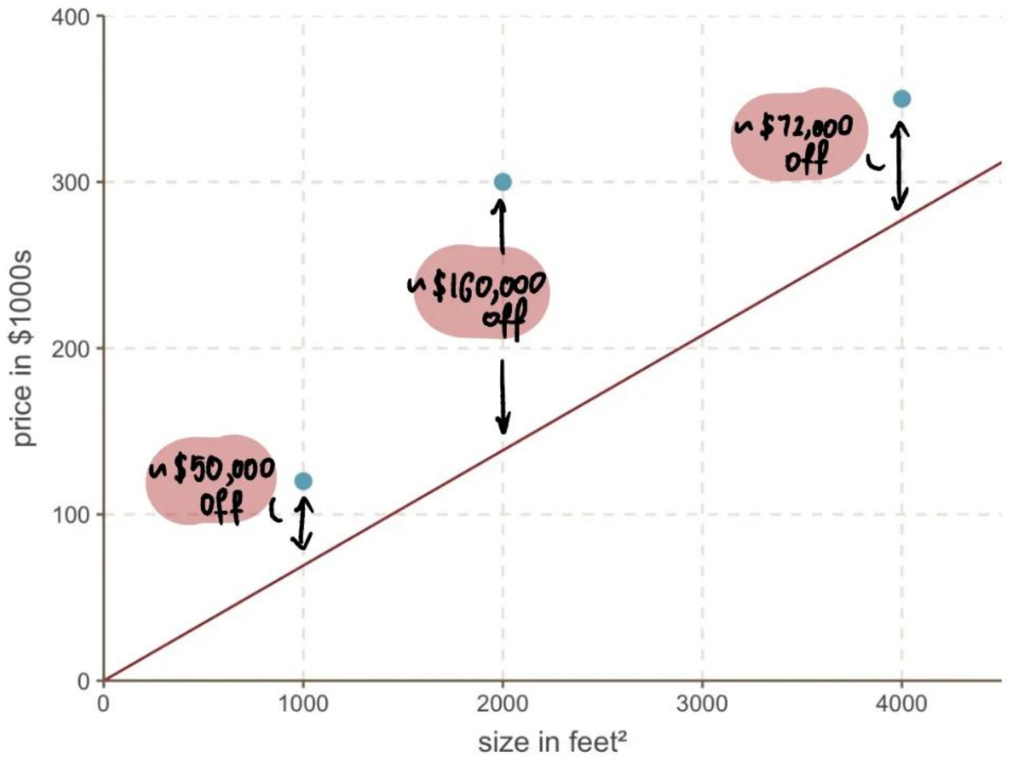


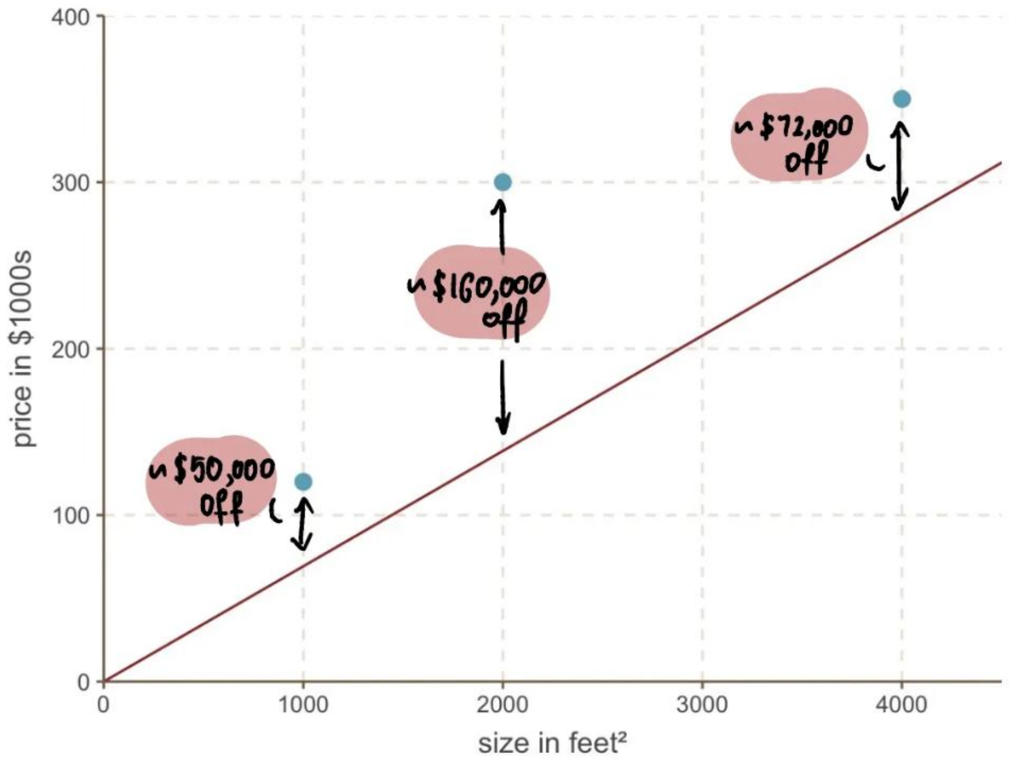



price in $1000s
400
300
200
100
0
0
1000
2000
3000
4000
size in feet²

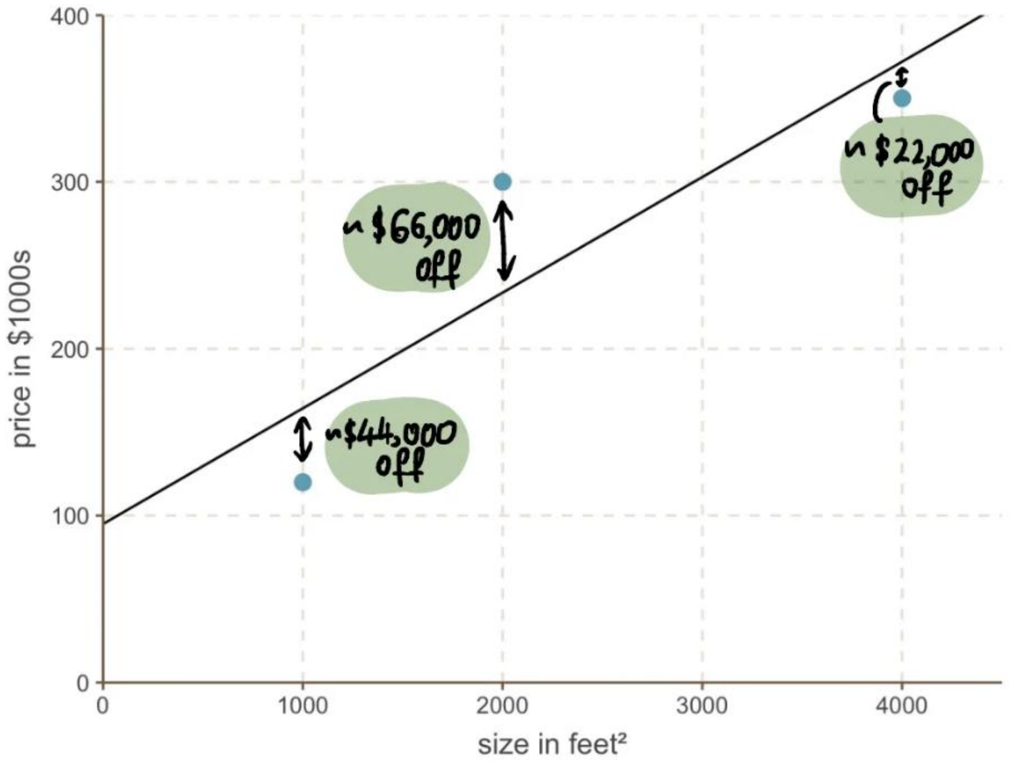

400
300
200
100
0
price in $1000s
0
1000
2000
3000
4000
size in feet²


actual price
prediction is ~$160,000 off
predicted price
price in $1000s
size in feet²
0
100
200
300
400
1000
2000
3000
4000


400
300
200
100
0
price in $1000s
0
1000
2000
3000
4000
size in feet²
~$50,000 off
~$160,000 off
~$72,000 off



400
300
200
100
0
price in $1000s
0
1000
2000
3000
4000
size in feet²
~$72,000 off
~$160,000 off
~$50,000 off



400
300
200
100
0
price in $1000s
0
1000
2000
3000
4000
size in feet²
~$22,000 off
~$66,000 off
~$44,000 off

② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y-\hat{y}|$$

④ calculate the average

③ sum of all the absolute differences

② the absolute difference between the actual and predicted house price

① predicted house price

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y - \hat{y}|$$

④ calculate the average

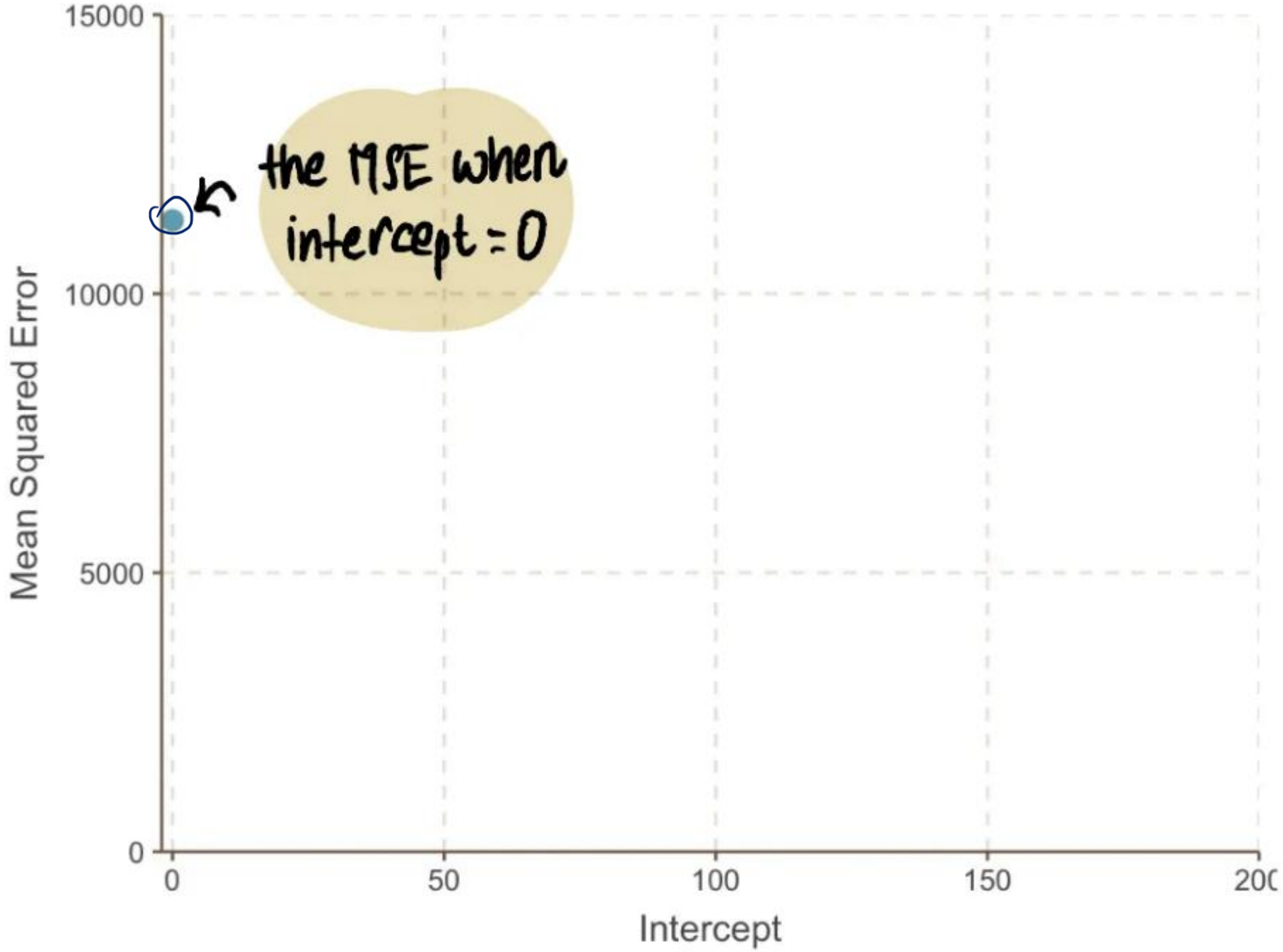
③ sum of all the absolute differences

actual house price

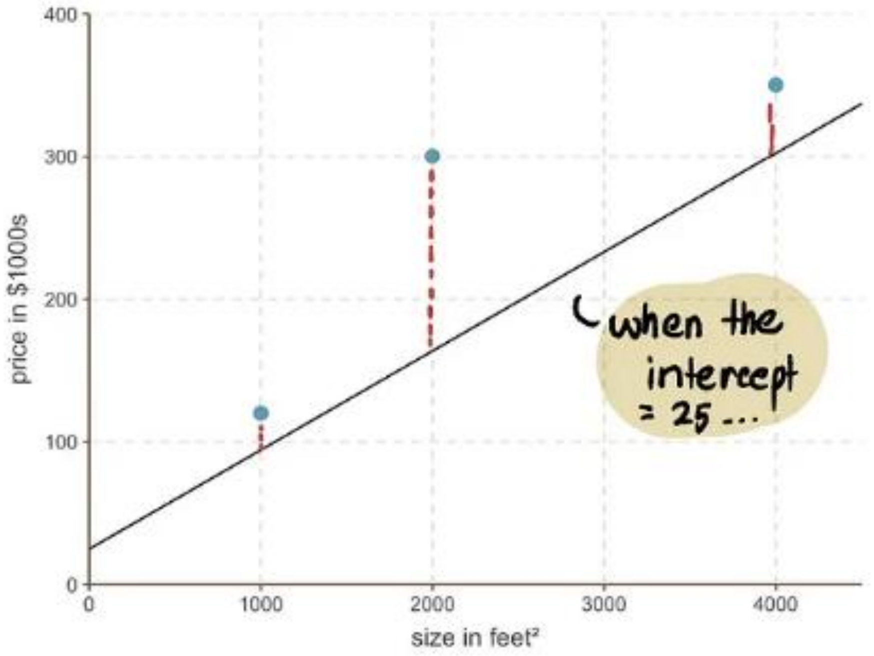
predicted house price when Intercept = 0

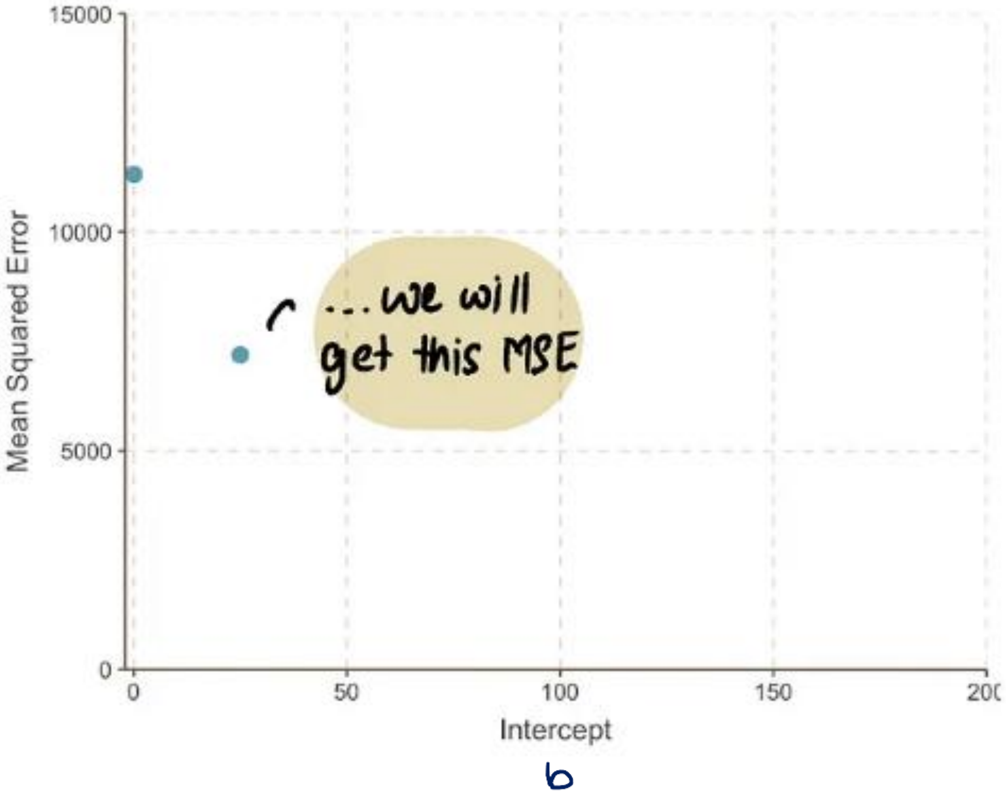
$$MSE = \frac{1}{3}\left[(120 - (\underbrace{0 + 0.069 \times 1000}))^2 + (300 - (0 + 0.069 \times 2000))^2 + (350 - (0 + 0.069 \times 4000))^2\right]$$

$$= 11313$$



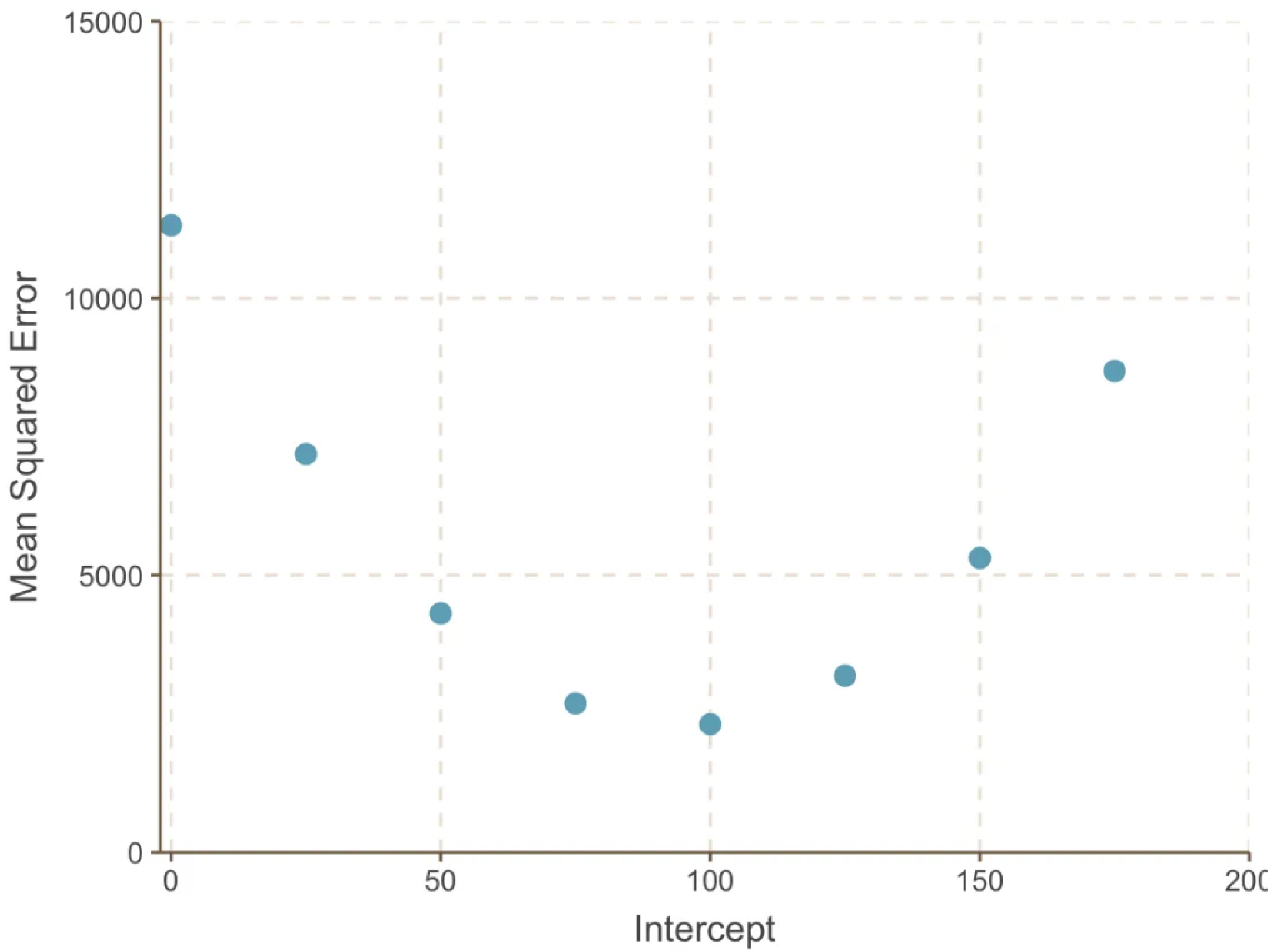
$y = ax + \underbrace{b}_{=0}$

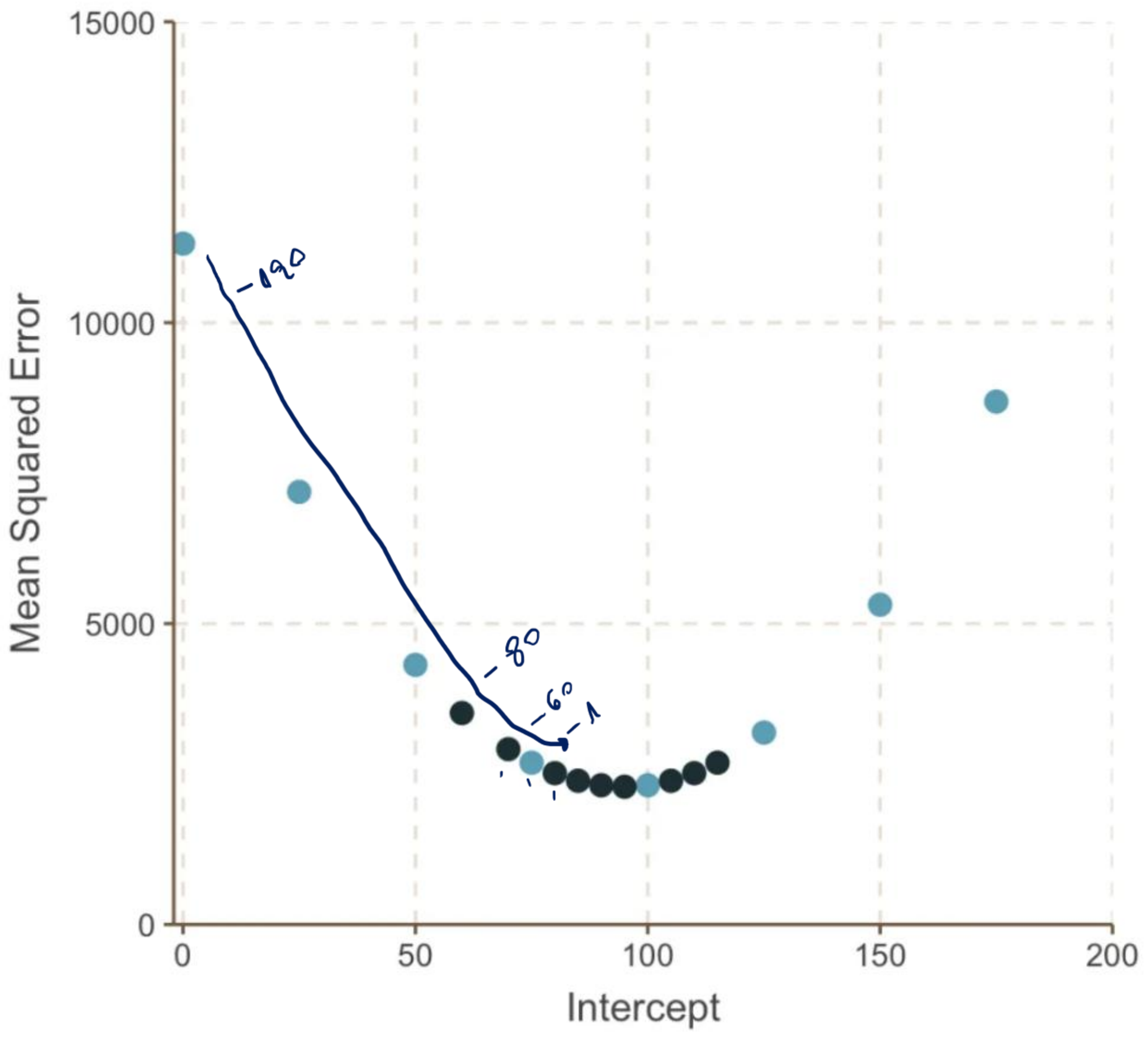


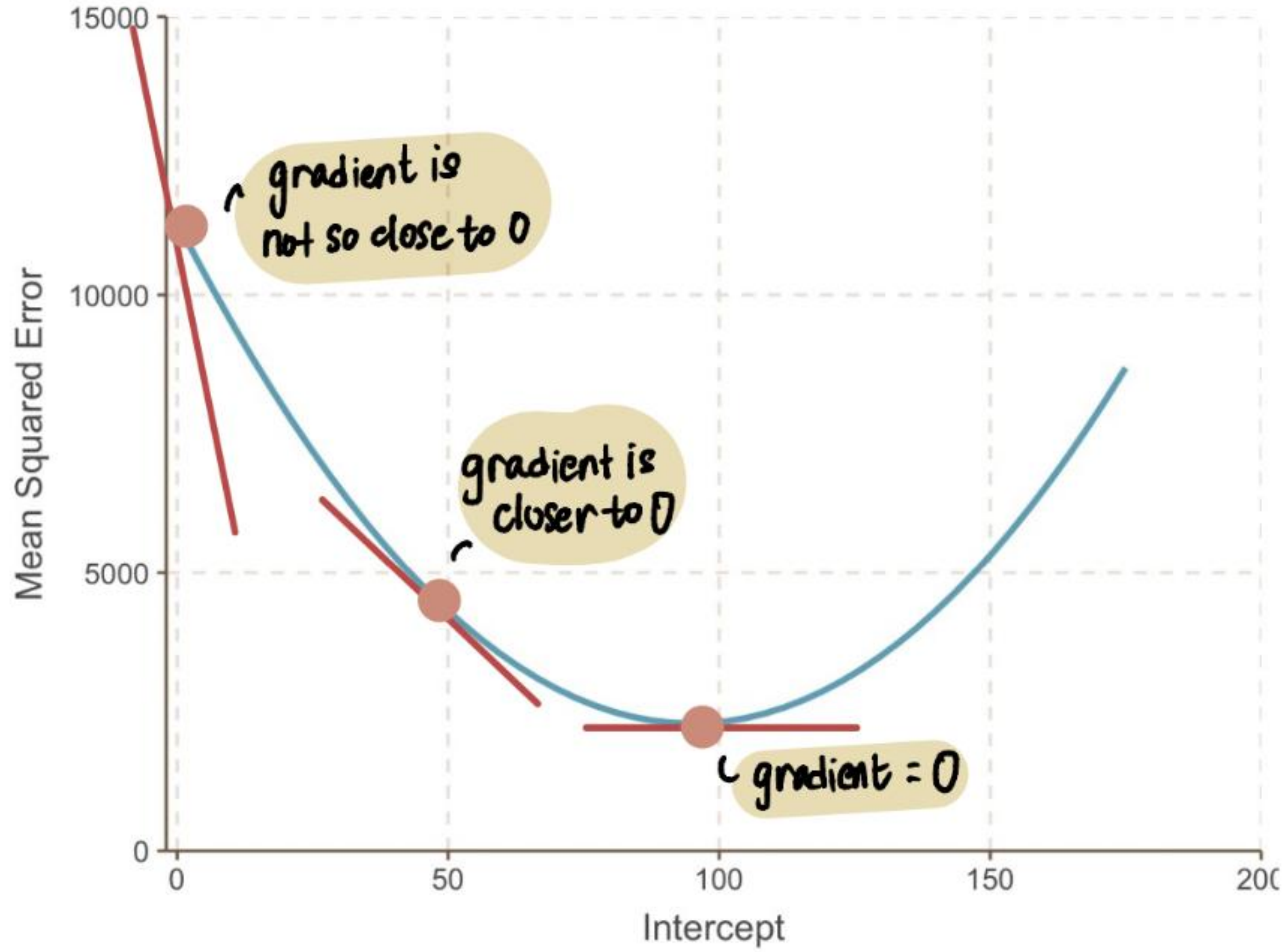


$$MSE = \frac{1}{3}\left[(120-(25+0.069\times 1000))^2 + (300-(25+0.069\times 2000))^2 + (350-(25+0.069\times 4000))^2\right]$$

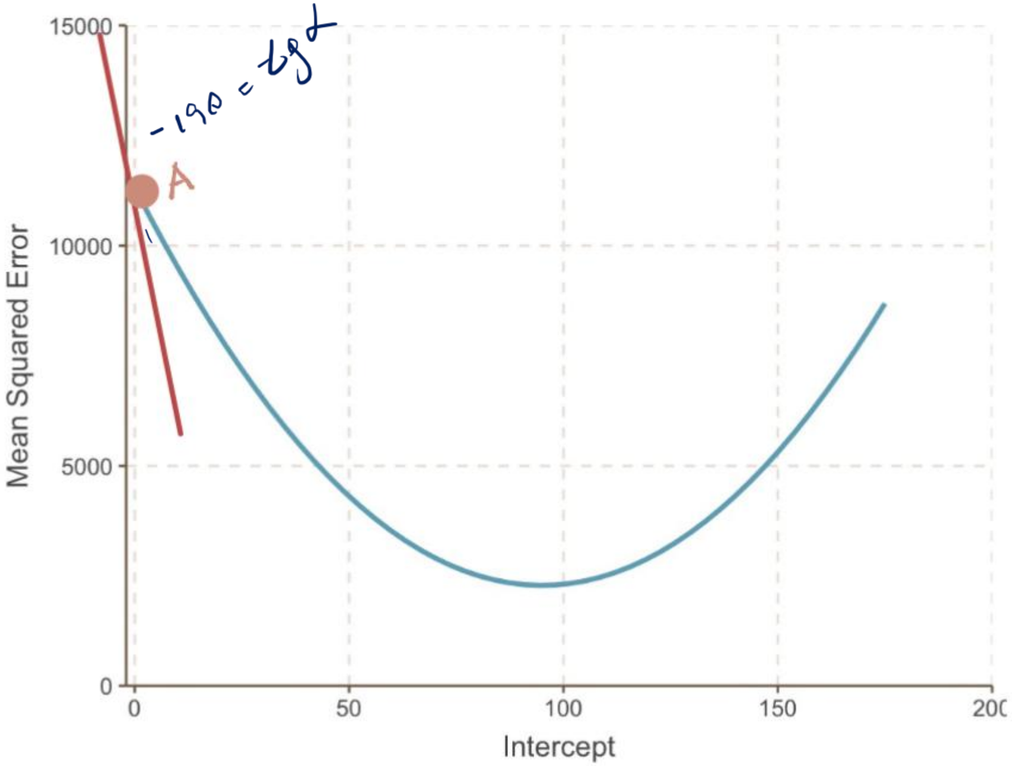
$$= 7188$$


15000
10000
5000
0
Mean Squared Error
0
50
100
150
200
Intercept


15000
10000
5000
0
Mean Squared Error
0
50
100
150
200
Intercept
-120
-80
-60
-1


15000
10000
5000
0
Mean Squared Error
0
50
100
150
200
Intercept
gradient is not so close to 0
gradient is closer to 0
gradient = 0

b=0



① derivative is denoted by 'd'

③ derivative at a specific value of the intercept

$$\frac{d\,MSE}{d\,intercept}\Bigg|_{intercept}$$

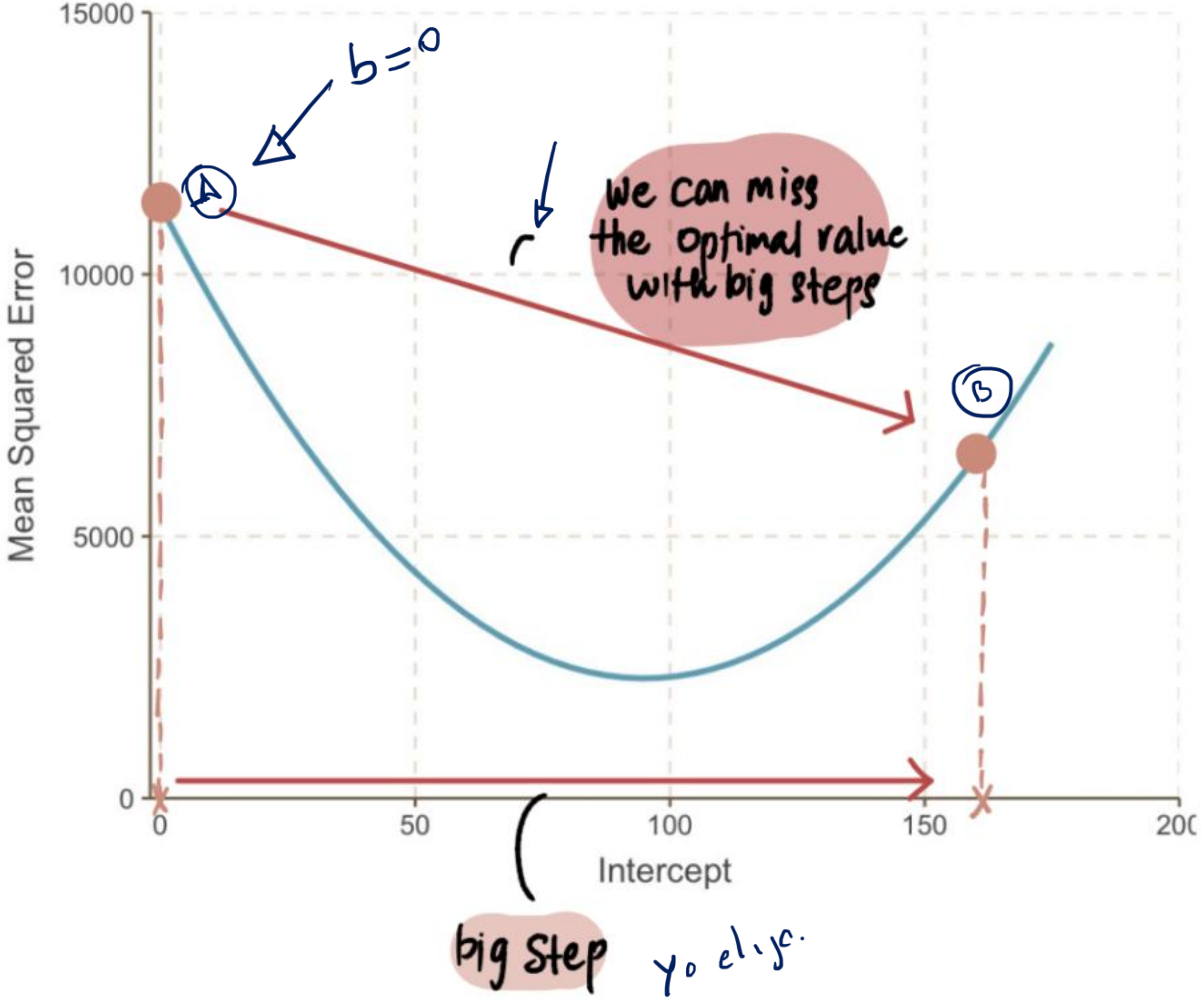
② derivative with respect to the intercept

$$\frac{dMSE}{d\,intercept} = \frac{d}{d\,intercept}\left[\frac{1}{3}\left[(120-(intercept+0.069\times1000))^2 + (300-(intercept+0.069\times2000))^2 + (350-(intercept+0.069\times4000))^2\right]\right]$$

$$= \frac{1}{3}\left[-2(120-(intercept+0.069\times1000)) - 2(300-(intercept+0.069\times2000)) - 2(350-(intercept+0.069\times4000))\right]$$

$$\left.\frac{dMSE}{d\,intercept}\right|_0 = \frac{1}{3}\left[-2(120-(0+0.069\times1000)) - 2(300-(0+0.069\times2000)) - 2(350-(0+0.069\times4000))\right]$$

$$= -190$$



Yo elijo.

calc

$$\text{Step Size} = \text{Learning Rate} \times \text{gradient}$$

learning rate

gradient when intercept = 0

$$\text{Step Size} = 0.1 \times -190$$
$$= -19$$

dirección

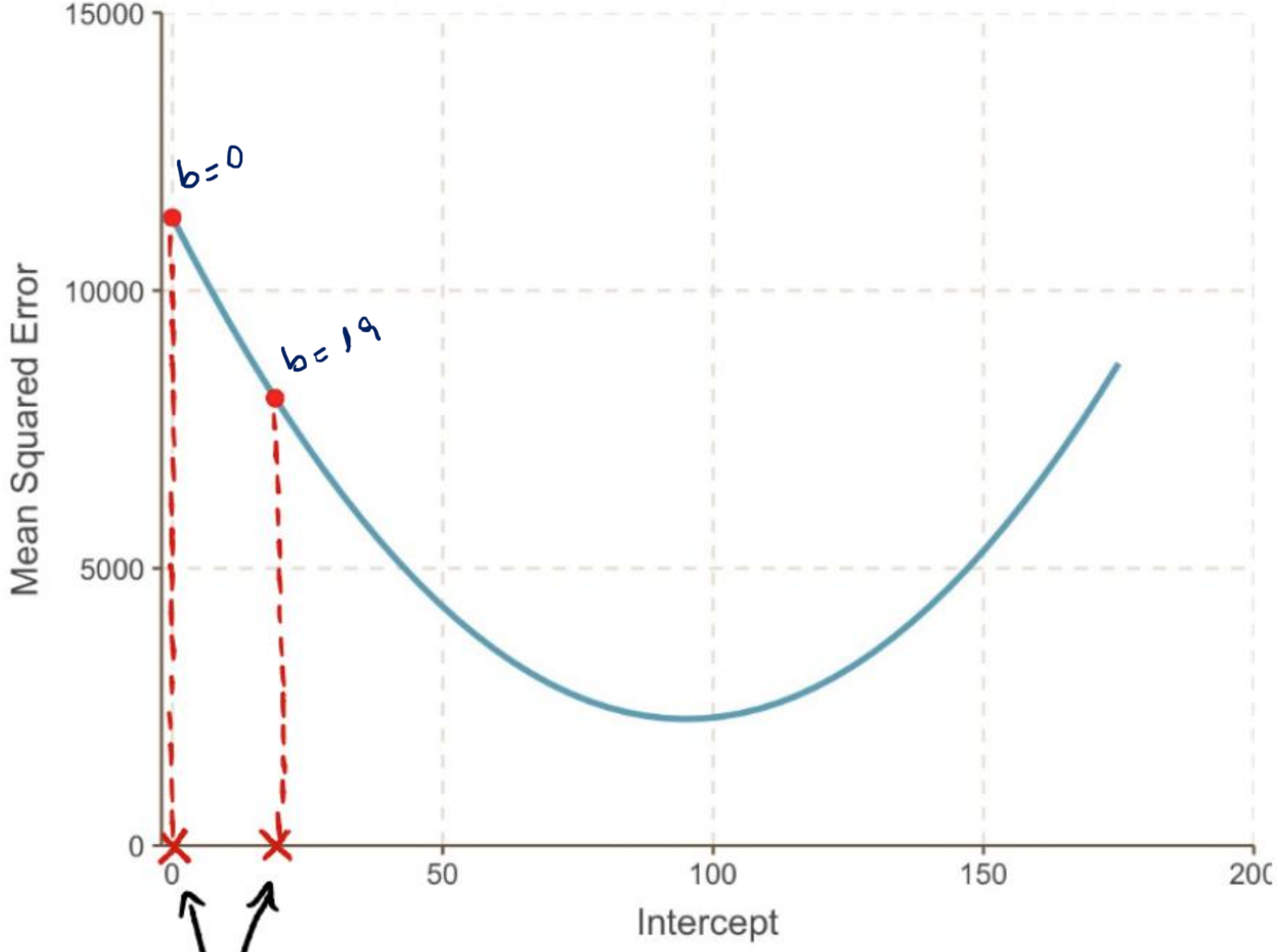
$$\text{New Intercept} = \text{Old Intercept} - \text{Step Size}$$

$$\boxed{b - r \cdot grad}$$

$$\text{New Intercept} = 0 - (-19)$$

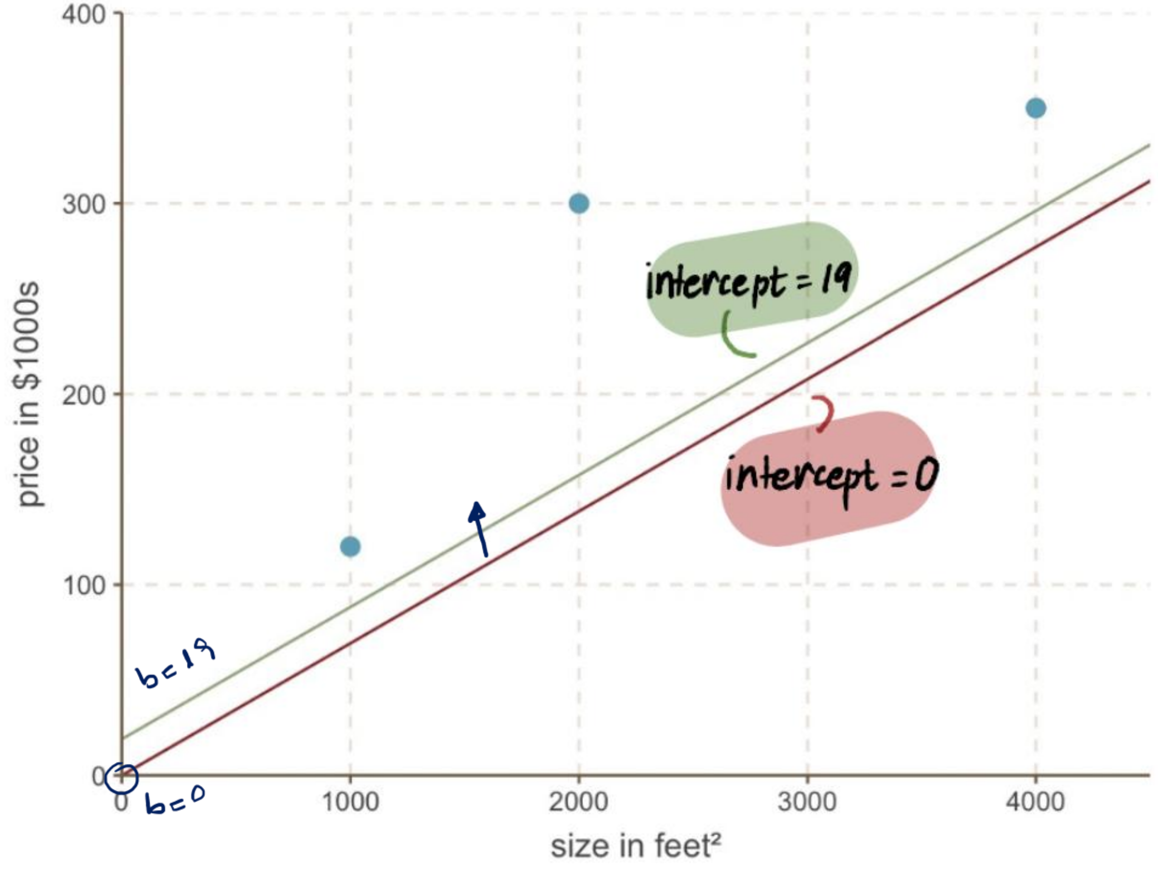
$$\boxed{b = 19}$$

$$W_{12} - r \cdot grad \Rightarrow \boxed{W_{12} - \text{Step Size}}$$



$$\text{New Intercept} = 0 - (-19)$$
$$= 19$$

in one big step, we moved much closer to our optimal value

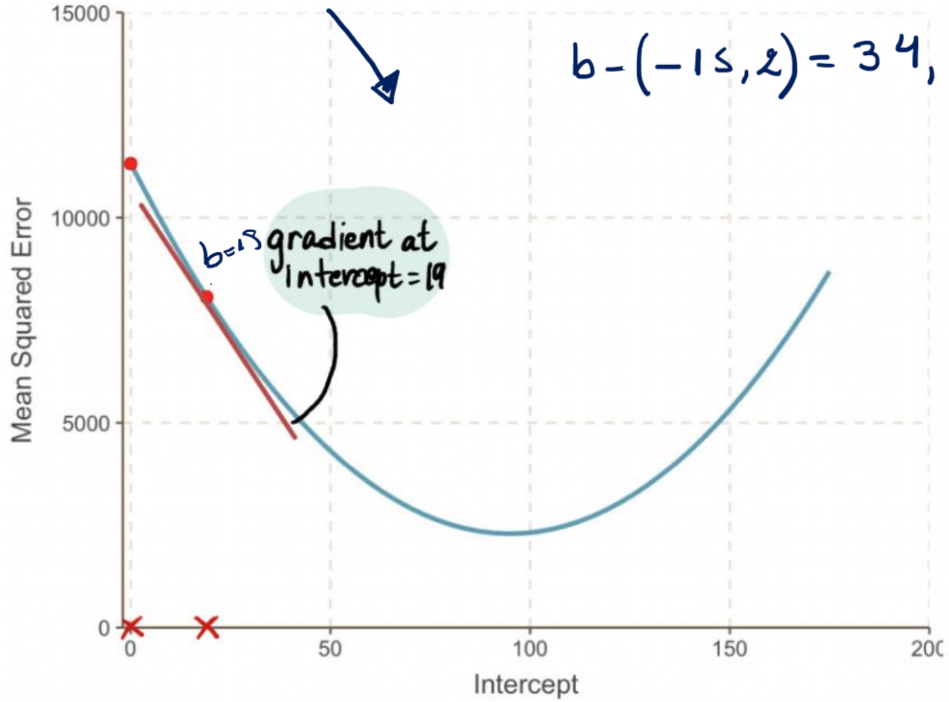


$$\left.\frac{dMSE}{d\ intercept}\right|_{19} = \frac{1}{3}\Big[-2(120-(19+0.069\times 1000)) -2(300-(19+0.069\times 2000)) -2(350-(19+0.069\times 4000))\Big]$$

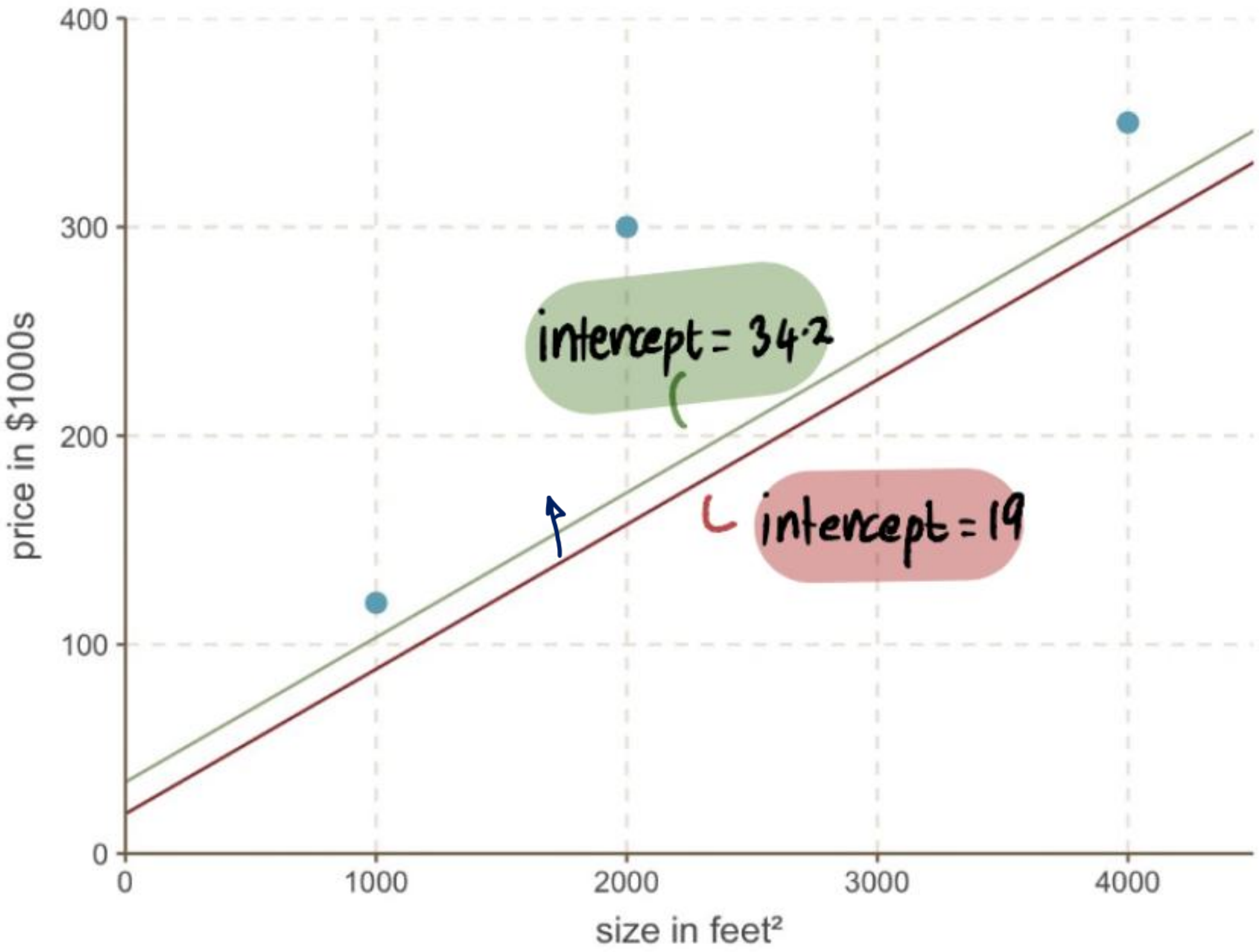
$$= -152$$

r.grad = 0,1 . (-152) = -15,2

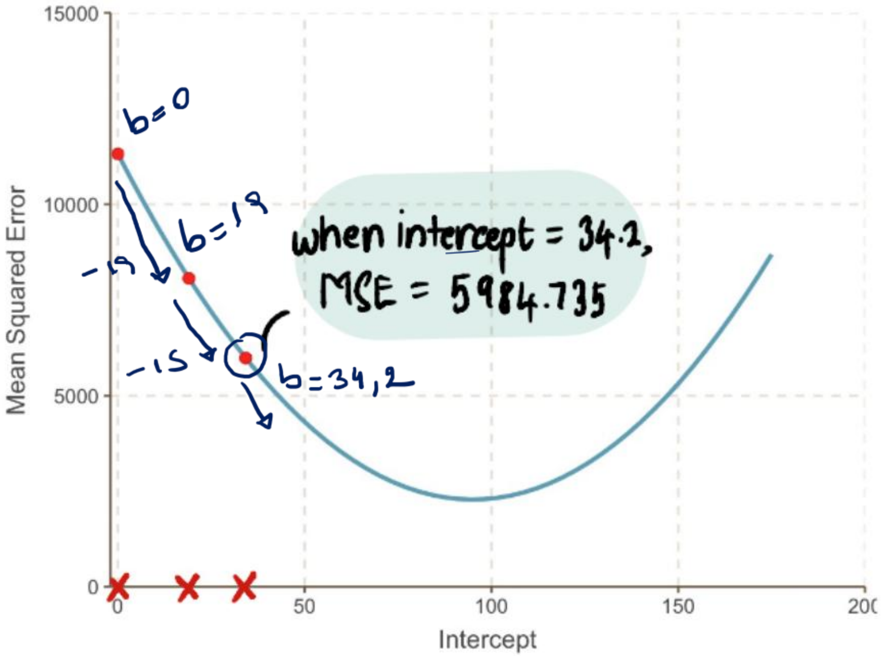
b - (-15,2) = 34,2

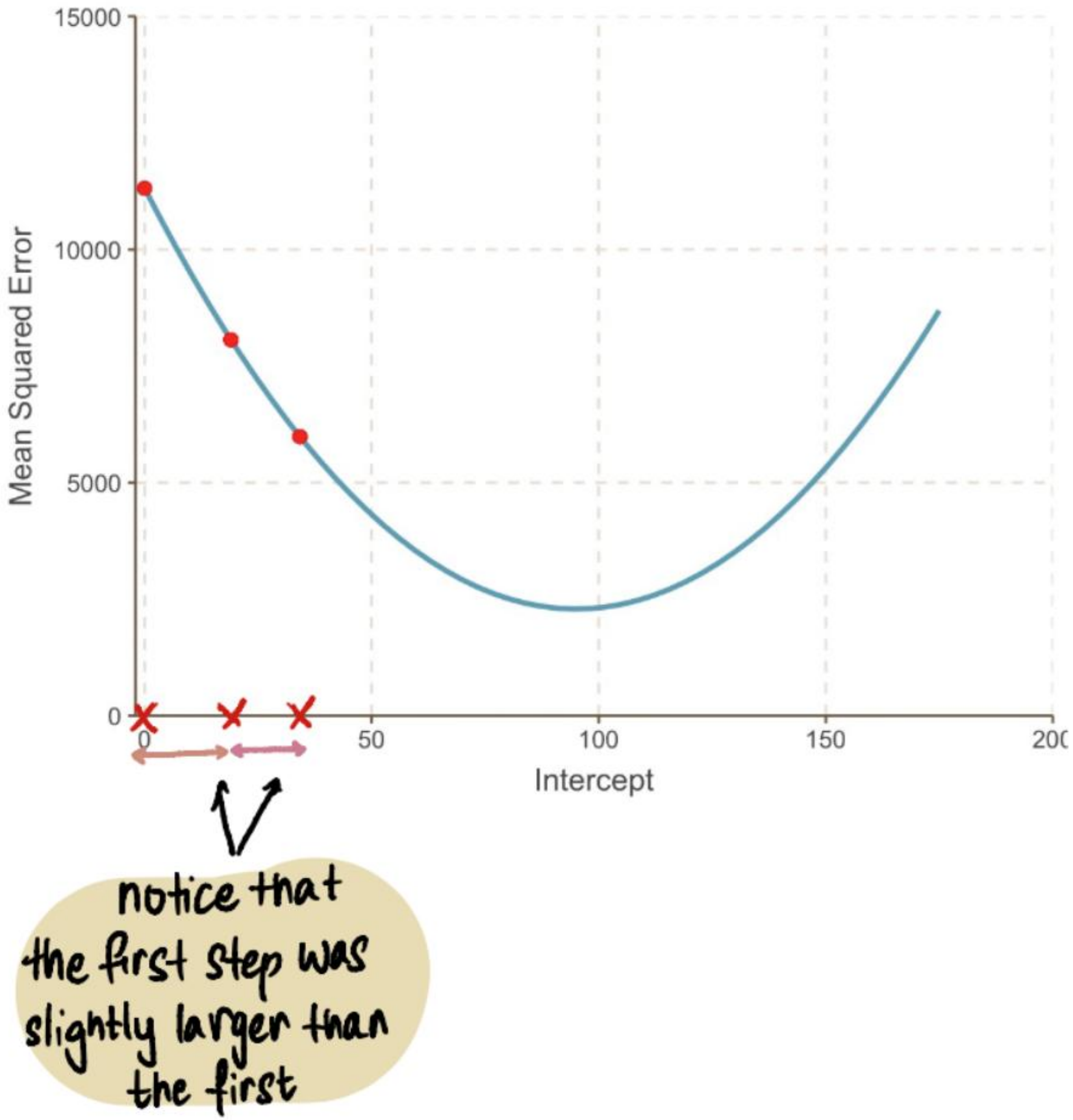


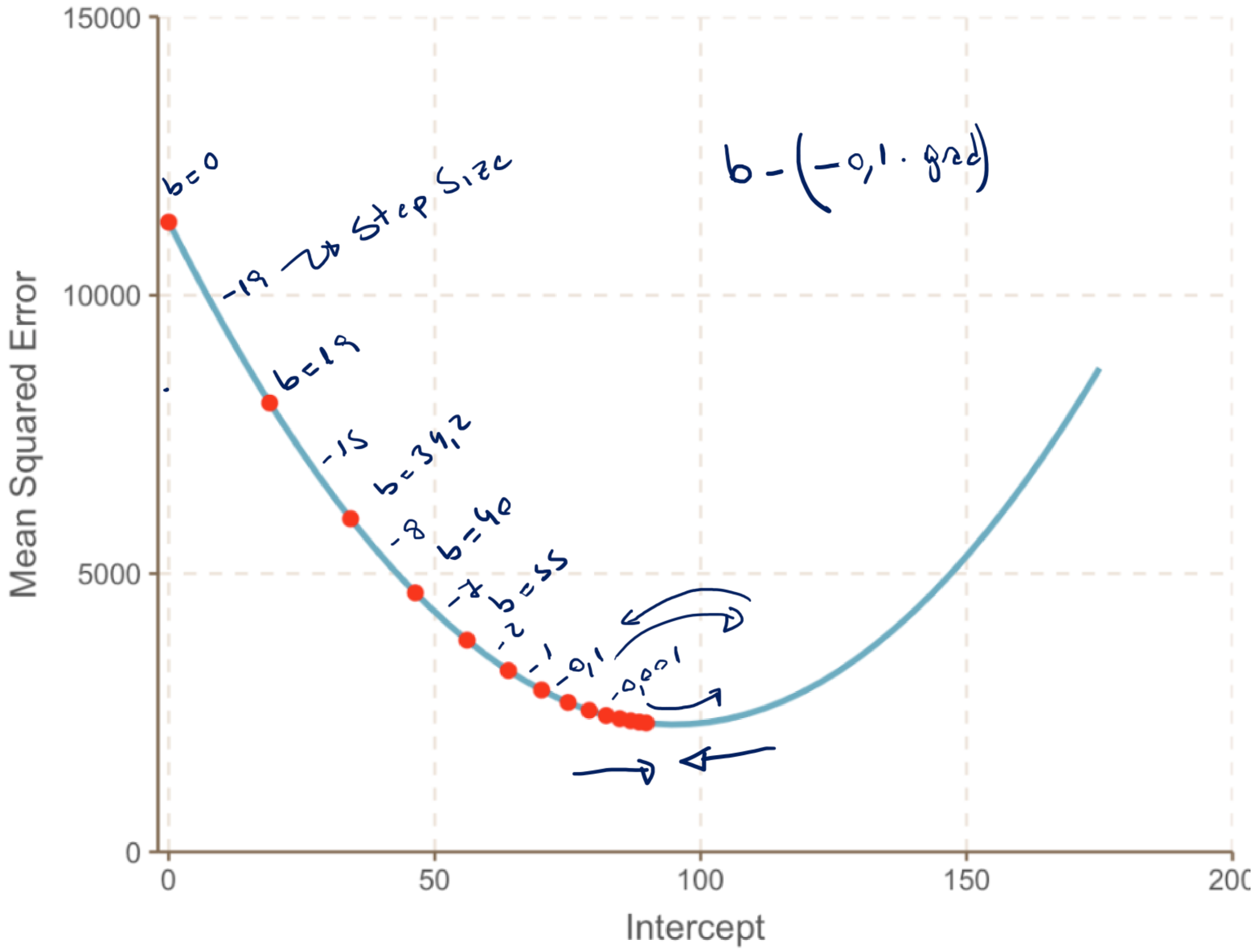
$$\text{Step Size} = \text{Gradient} \times 0.1 = -152 \times 0.1 = -15.2$$



$$\text{New Intercept} = \text{Old Intercept} - \text{Step Size} = 19 - (-15.2) = 34.2$$




15000
10000
5000
0
Mean Squared Error
0
50
100
150
200
Intercept
notice that the first step was slightly larger than the first



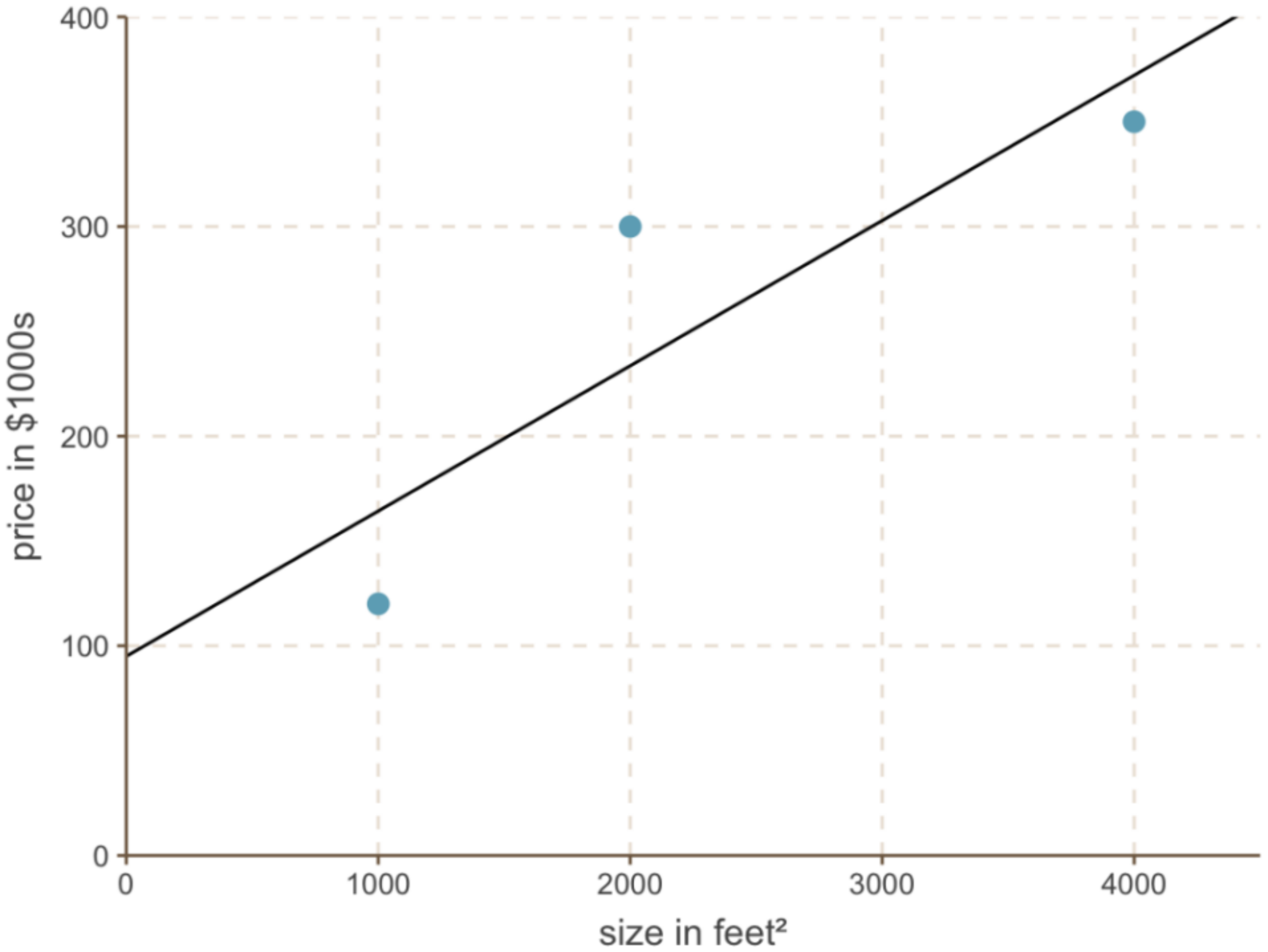
https://medium.com/

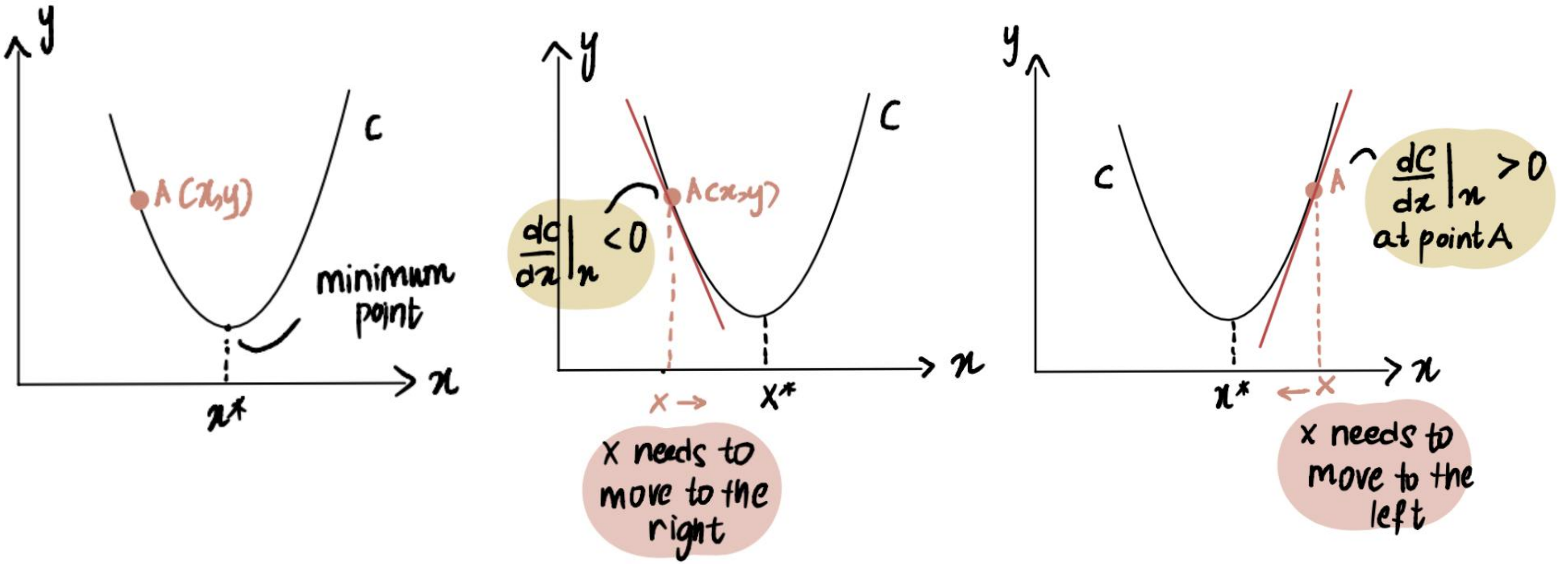
predicted house price

$$\hat{y} = 95 + 0.069 \times size$$

optimal intercept value we found

$$\hat{y} = 95 + 0.069 \times 2400$$
$$= 260.6$$

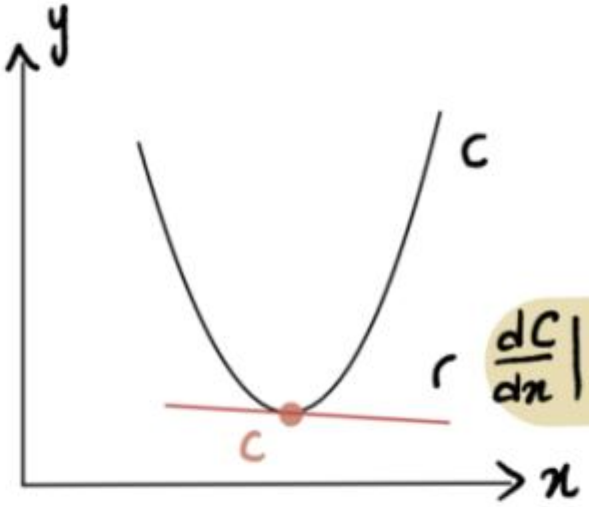




https://medium.com/

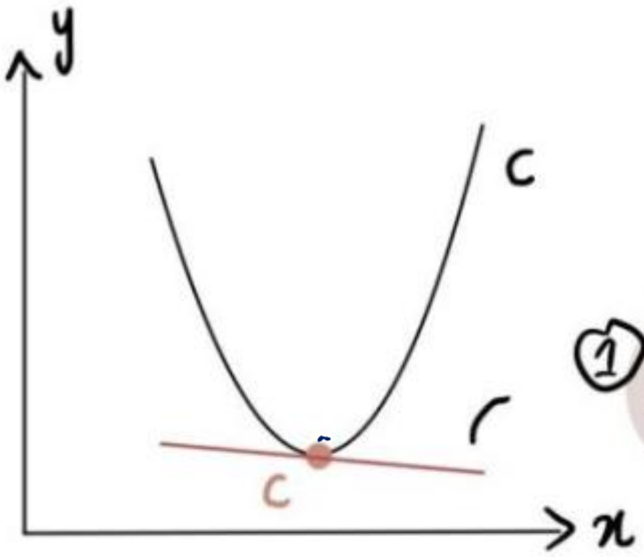
Step Size = Learning Rate x Gradient

Step size will be close to 0 when gradient is close to 0



$\left.\frac{dC}{dx}\right|_x$ is close to 0

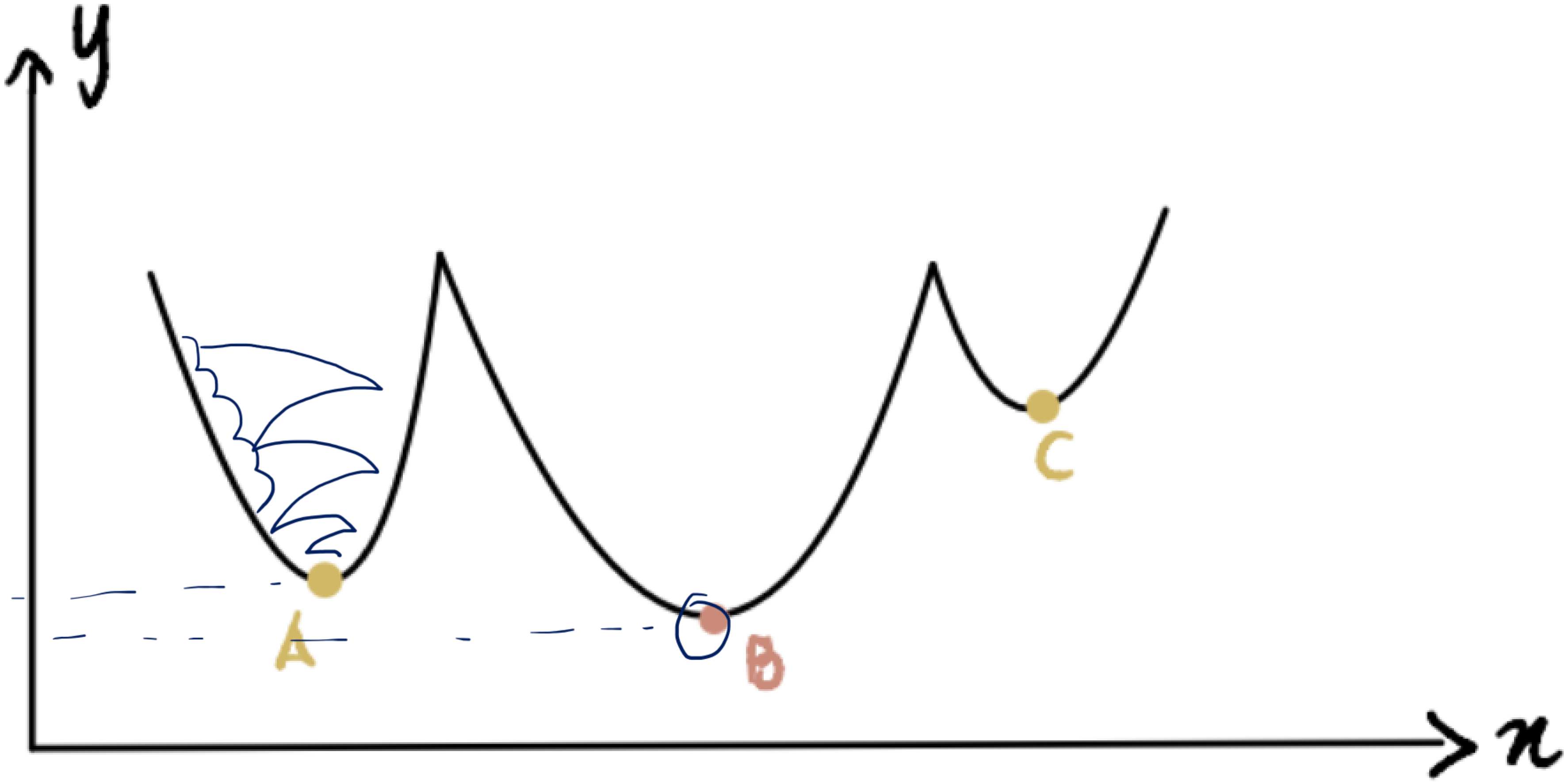
https://medium.com/

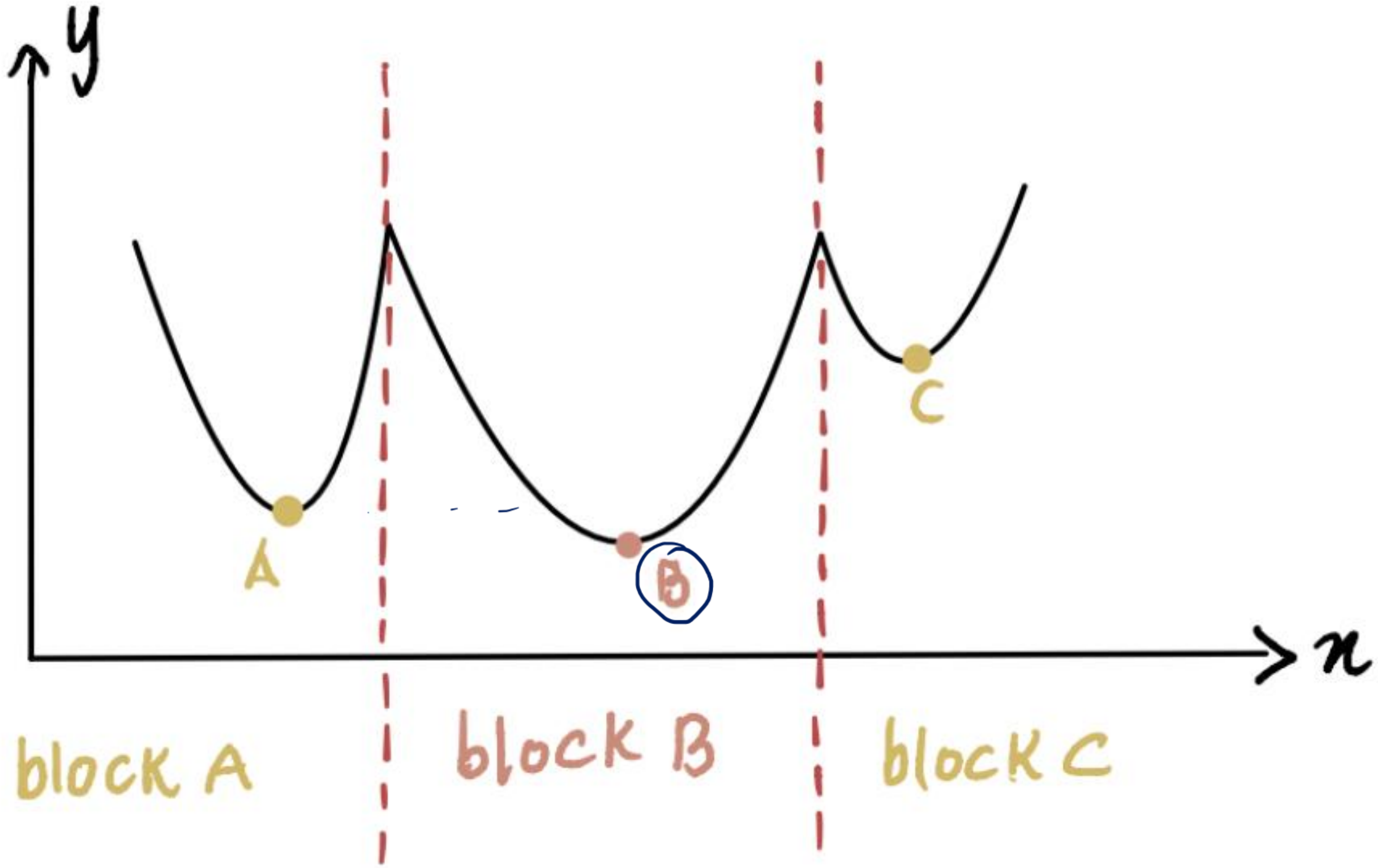


① if the gradient here = 0.009

② Step Size = 0.009 X 0.1
= 0.0009

③ 0.0009 < 0.001, so Gradient Descent will stop


y
x
A
B
C



y
x
A
B
C
block A
block B
block C

this symbol denotes partial derivative

$$\underbrace{\frac{\partial C(x_0,x_1)}{\partial x_0}}_{\text{partial derivative with respect to } x_0}, \quad \underbrace{\frac{\partial C(x_0,x_1)}{\partial x_1}}_{\text{partial derivative with respect to } x_1}$$

$$x_0 = x_0 - \overbrace{\alpha \times \frac{\partial C(x_0,x_1)}{\partial x_0}}^{\text{Step size for } x_0}$$

$$\overbrace{x_1}^{w,b} = x_1 - \overbrace{\alpha}^{\eta} \times \underbrace{\frac{\overbrace{\partial C(x_0,x_1)}^{\text{grad}}}{\partial x_1}}_{\text{Step size for } x_1} \quad \nabla$$